

Management Consulting Case Interviews

Visit our website: <http://www.consultingcase101.com>

More consulting resources at: <http://www.consultingcase101.com/resources/>

Join Consulting Forum: <http://www.consultingcase101.com/forum/>

Our Blog: <http://www.consultingcase101.com/blog/>

Subscribe via RSS Feed: <http://www.consultingcase101.com/feed/>

Subscribe via Feedburner: <http://feeds.feedburner.com/consultingcase101>

Subscribe via Amazon Kindle: <http://www.amazon.com/Management-Consulting-Case-Interviews/dp/B0041HXS3Y/>

Follow us on Twitter: <http://twitter.com/consultingcase>

Follow us on Facebook: <http://www.facebook.com/pages/Management-Consulting-Case-Interviews/140533409321335/>



YALE SOM
CONSULTING CLUB

Week 1 Club Meeting

September 11, 2008

These written materials are not complete without accompanying verbal comments

Educating Leaders for Business and Society

 **YALE SCHOOL OF MANAGEMENT**

Agenda

- ▶ Club Administrative Details
- ▶ Case Demonstration
- ▶ Overview of Case Framing
- ▶ Case-Framing Exercise

Club attendance policy

- ▶ Attendance at events is not mandatory
- ▶ Recommend students sign-in during meetings and Consulting Club events
 - Point system
- ▶ Casing preparation & behavioral practice priority given to students with highest point count

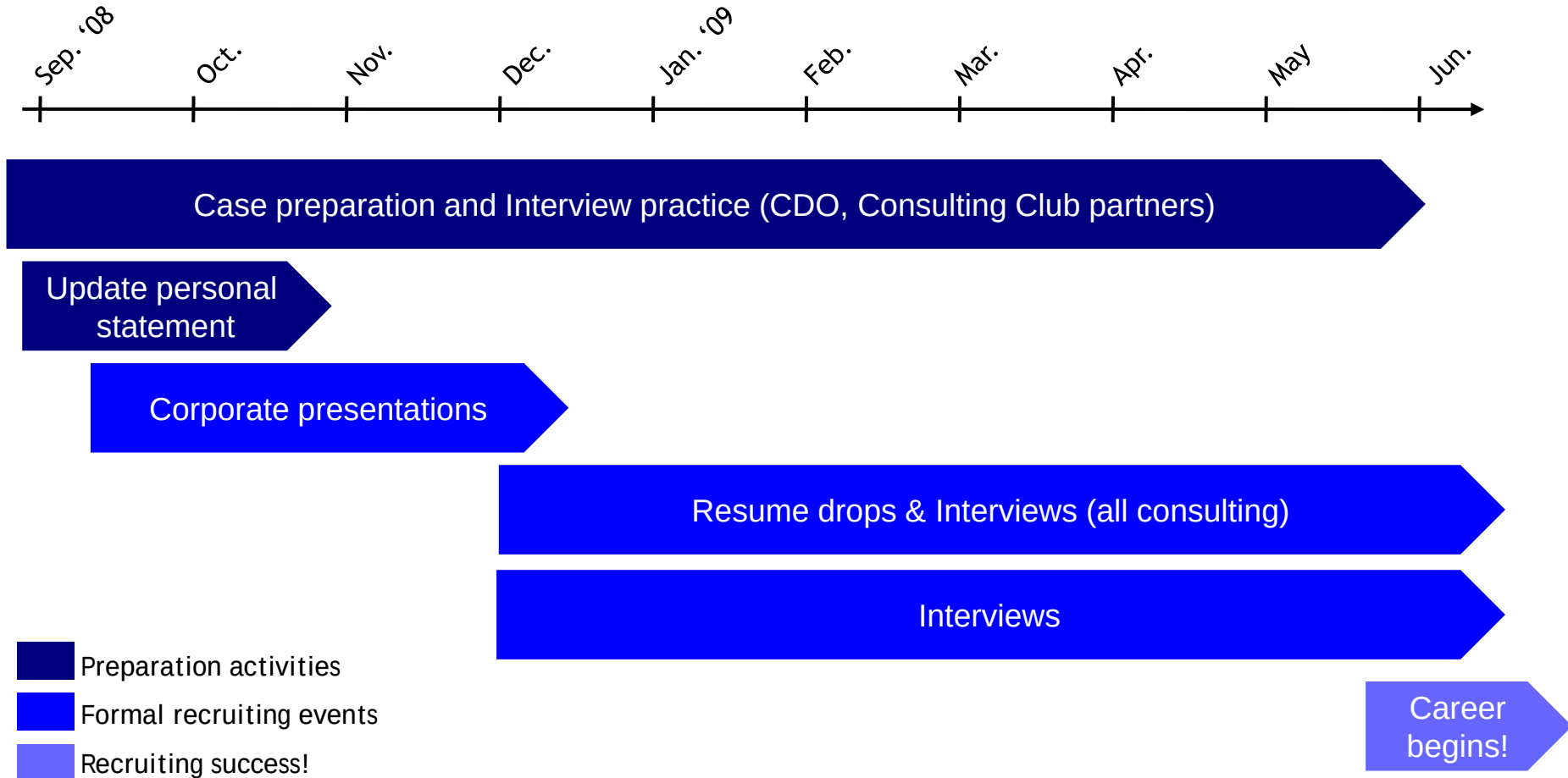
Preparation FAQ

- ▶ How many cases should you do?
 - Depends on the person and prior experience, but aim for **at least 40 cases**

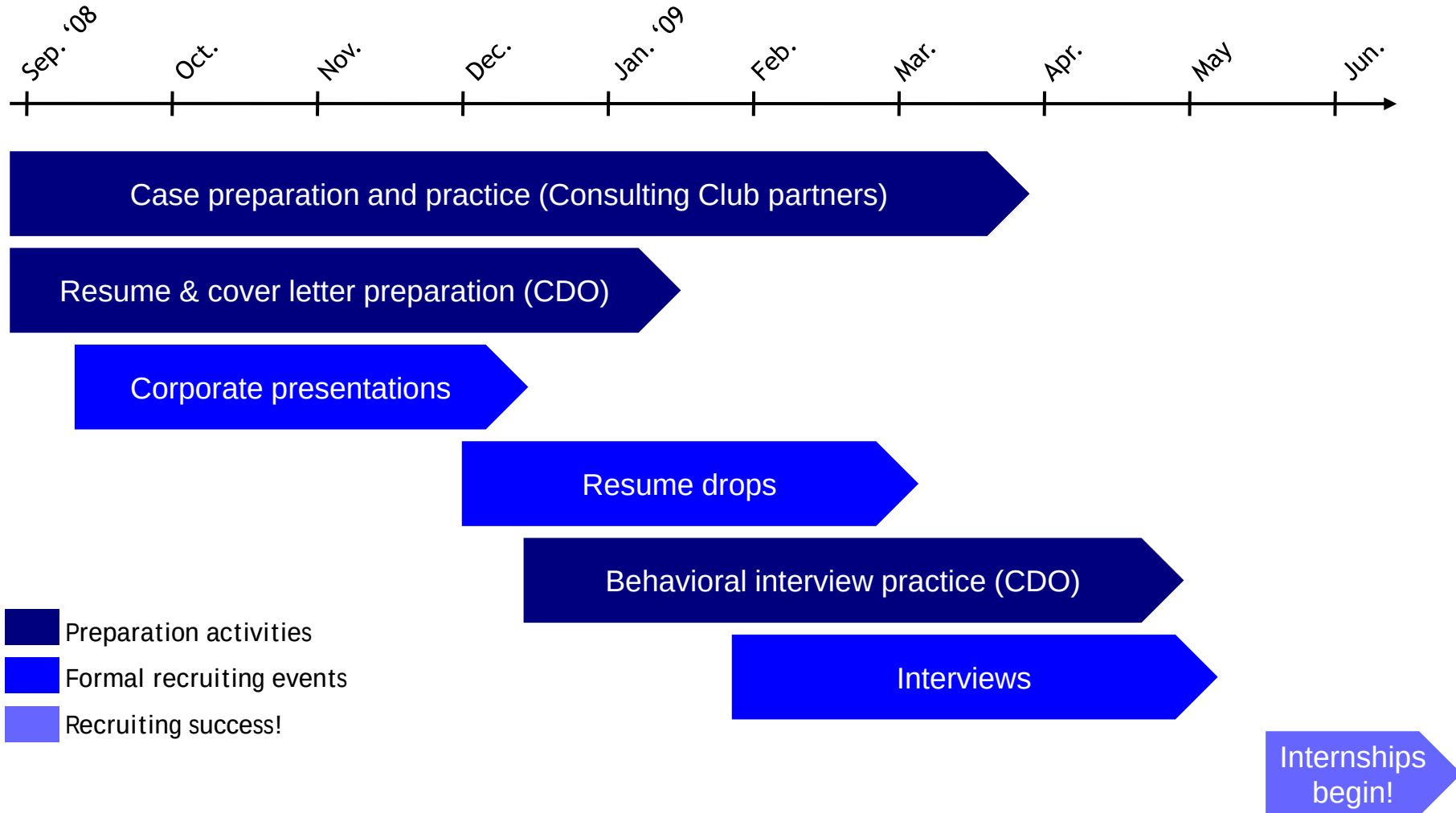
- ▶ When should I start crafting my story?
 - Second Years: **NOW**
 - First Years: **NOW**

- ▶ When should I start actively preparing for the behavioral component?
 - Second Years: **NOW**
 - First Years: **After Fall-1 exams**

Example Second-Year Recruiting Process for Consulting



Example First-Year Recruiting Process for Consulting



Consulting Club Curriculum Overview - Fall 2009

WEEKLY TOPICS

- ▶ *September 11*: Framing the problem
- ▶ *September 18*: Review of different types of cases and frameworks. Choosing 'right' framework or building your own
- ▶ *September 25*: Commonly tested Math concepts . Taking the notes during the interview
- ▶ *October 2*: Wrapping up the interview: Closing the case and identifying insights. Review of the common academic concepts . Balance of "other issues" and to-the-point analysis
- ▶ *October 9*: Maintaining creativity in analyzing standard problems. Many facets of one issue – thinking out of the box.
- ▶ *October 16*: Guest case workshop with BCG. Case practice

Consulting Club Curriculum Overview contd.

- ▶ *October 30:* Using the (integrated!) core curriculum as a framework . Review of core classes
- ▶ *November 6:* Data. What data to look for. How to analyze and use the data. Analysis of graphs. Review of common data by case type
- ▶ *November 13:* Guest case workshop (TBD)
- ▶ *November 20:* Behavioral interview.
- ▶ *December 4:* Problem Framing and Math revisited
- ▶ *December 11:* Final Overview and covering the 'weak spots'

THE TOPICS CAN BE ALTERED OR ADDED DEPENDING ON YOUR COMMENTS

Casing Lists for 1st and 2nd Years

Second Years

- ▶ Saturday round-robin group case practice session
- ▶ Sign up on our Google Document
- ▶ Refer to the Google Document to find names for scheduling individual case practice sessions

First Years

- ▶ Organized case list to come in a few weeks!
- ▶ Don't be shy to ask classmates to set up a time to give a case to one another
- ▶ Consulting Club Superweek of Casing – Coming in January!

Agenda

- ▶ Club Administrative Details

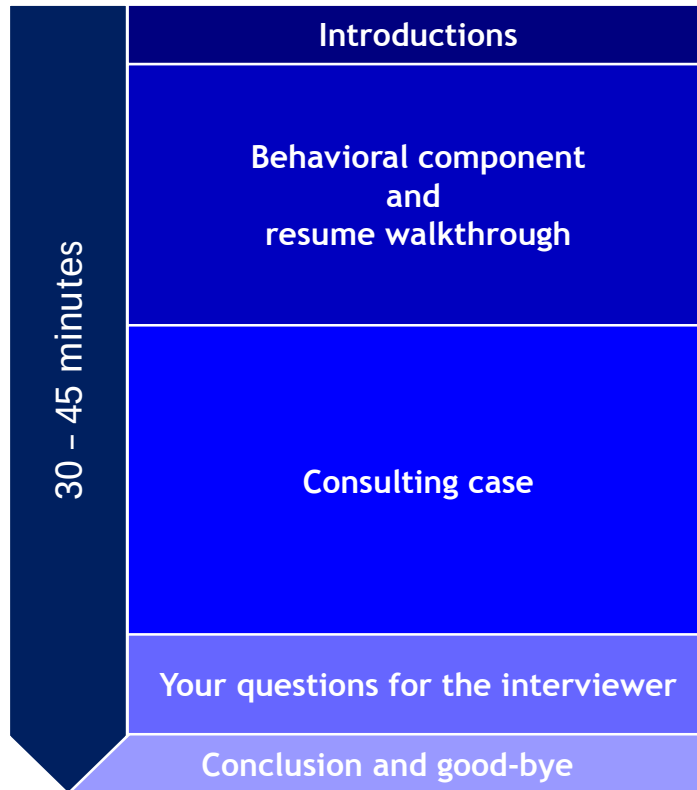
- ▶ Case Demonstration

- ▶ Overview of Case Framing

- ▶ Case-Framing Exercise

The Consulting Interview

The Interview



- ▶ These are the two most important sections of the interview
- ▶ Many of the same skills are tested in each section - especially the ability to speak clearly, logically, and confidently
- ▶ The case is your chance to demonstrate the skills you talk about during the behavioral component

Behavioral Issues

- ▶ Speaking with confidence
- ▶ Looking the part
- ▶ Preparing for the interview
- ▶ Conscious of body language
- ▶ Building rapport
- ▶ Staying positive
- ▶ Mastering the Q & A

Case Demonstration

What to watch for:

- ▶ Demonstration of **good listening** skills by the interviewee
- ▶ Logical framework for **approaching** the problem
- ▶ Logical approach to **solving** the problem
- ▶ Making frequent **eye contact** with the case-giver
- ▶ Calm, **relaxed demeanor** throughout the case
- ▶ Talking through the process of solving the case – **sharing the thought process**
- ▶ Delivering a **concise final recommendation**

The Demonstration

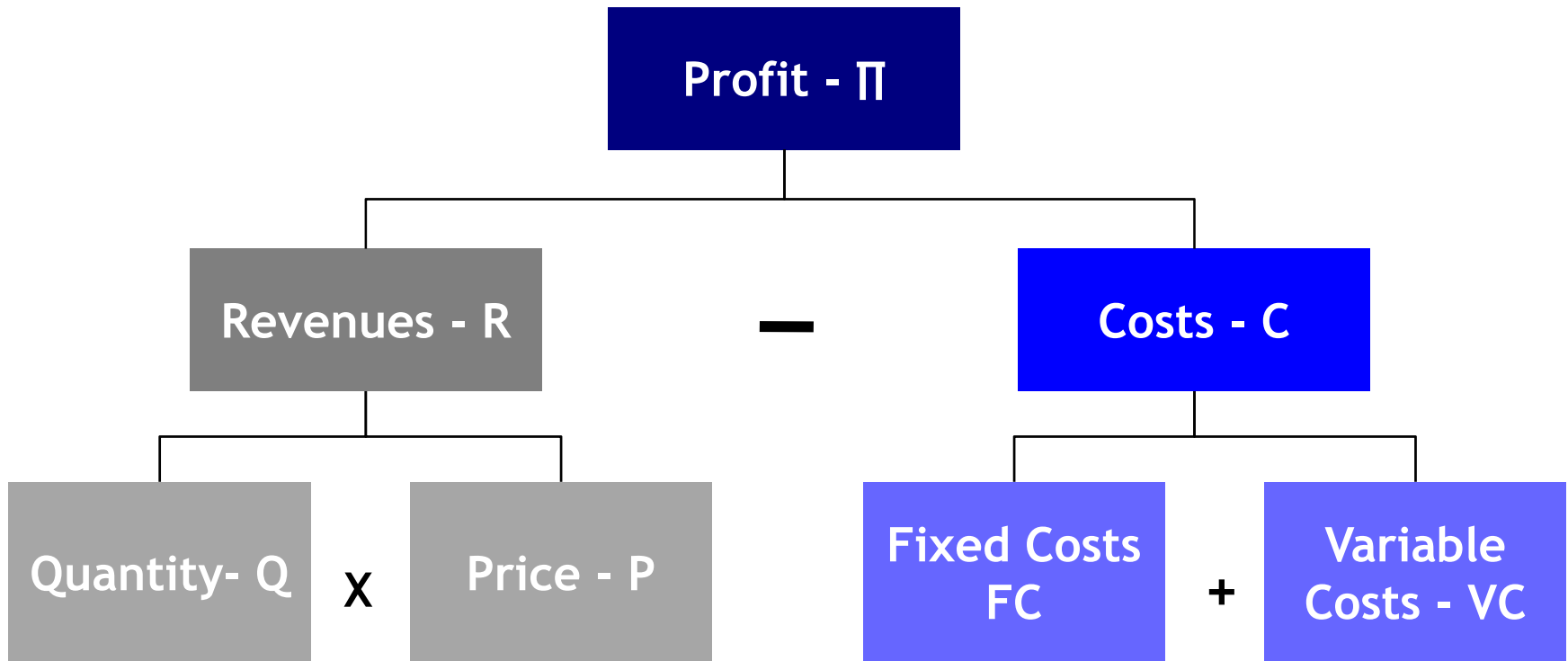
Agenda

- ▶ Club Administrative Details
- ▶ Case Demonstration
- ▶ Overview of Case Framing
- ▶ Case-Framing Exercise

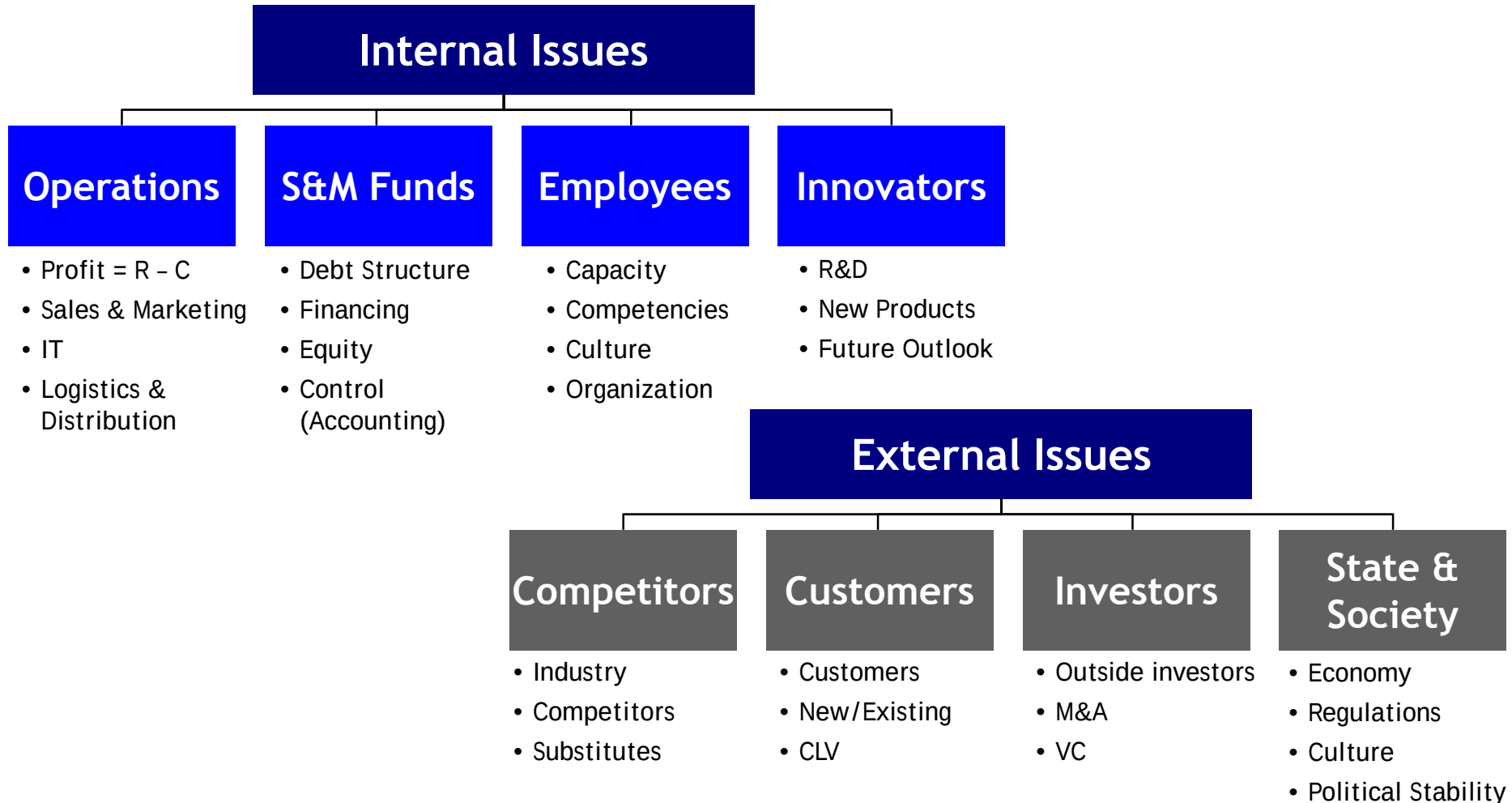
The Purpose of Frameworks

- ▶ Organize thoughts
- ▶ Bucket issues so they can be dealt with methodically and one-at-a-time
- ▶ Demonstrate that you can think logically, like a consultant
- ▶ Establish levels of abstractions, and work down as you solve the case
 - 30,000 feet
 - 10,000 feet
 - 1,000 feet
 - ...
- ▶ Frameworks help you track your progress when solving a case

Most-used Framework: The Profit Tree



The SOM Curriculum - A Consulting Framework



Creative Thinking

The SOM Core Curriculum does a good job teaching us to be broad, creative thinkers. However, there is a time and place for creativity, and practical judgment should be used

► GREEN INITIATIVES

- Who is the firm interviewing you?
- Who is the client in the case?
- What is the context of the case problem?

► “Other Issues” to explore in the case (State-and-Society stuff)

- Ask and “deal-killer” questions up front
- Focus on the core issues of the case first
- Use good judgment when discussing these ideas – be intelligently creative

► If it’s **illegal or unethical** – JUST STOP!! Don’t go there!

Agenda

- ▶ Club Administrative Details
- ▶ Case Demonstration
- ▶ Overview of Case Framing
- ▶ Case-Framing Exercise

Case Framing Exercise

Instructions

- ▶ Teams of 3 people
- ▶ Each person is given a case question
- ▶ Each person practices framing his/her approach to solving the case; THEN, verbally articulating the approach
- ▶ After all three team members have framed a case, go back and talk about the following:
 - What other issues could be covered?
 - What issues are most important?
 - What issues are less important?
 - What are your hypotheses?



YALE SOM
CONSULTING CLUB

2008 Consulting Club

Week 2

Overview of Case Framing

September 18, 2008

These written materials are not complete without accompanying verbal comments

Agenda

- ▶ Types of Cases
- ▶ Framing Strategy
- ▶ Common Reference Frameworks
- ▶ Applying Your Framework - Group Exercise

Overview of Basic Case Types

- ▶ Improving profitability
- ▶ Entering a market
- ▶ Developing a new product
- ▶ Estimating a market size
- ▶ Acquiring a company
- ▶ Analyzing an industry

Case Type: Improving Profitability

Improving Profitability		
General issue	Concepts often tested	Key solution elements
Your client is either suffering from a decline in profits, or simply wants to find a way to improve their profits *This is the most common type of case	• Cost-benefit analysis • Understanding the elements of revenues and costs • Ability to pinpoint a problem and suggest a solution	• Profit tree • Identification of fixed versus variable costs • Evaluation of typical methods for driving revenues

Case Type: Entering a Market

Market Entry		
General issue	Concepts often tested	Key solution elements
Your client wants to sell their products or services to a market which they have not served before	• Understanding of typical ways to enter a market (buy a company, joint-venture, etc.) • Understanding of customers, competitors, and industries	• Industry analysis • Market share and profit expectations • Competitive response expectations • Evaluation of entry options

Case Type: Developing a New Product

New Product		
General issue	Concepts often tested	Key solution elements
Your client is considering developing/launching a new product and wants your opinion on the matter	• Ability to evaluate the attractiveness of a new product • Profitability analysis, incl. price-setting • Understanding of customers, competitors, and industries	• Industry/competitor analysis • Pricing analysis • Market size estimate • Evaluation of new product's impact on the company (cost-benefit analysis)

Case Type: Estimating Market Size

Market Sizing		
General issue	Concepts often tested	Key solution elements
You need to estimate the number of people constituting the market for a particular product or service. Often appears as a case inside a case	• Ability to make assumptions • Ability to make reasonable judgments • Ability to think broadly and creatively • Basic math	• Plausible, round numbers • Intelligent assumptions • Efficient summary of data and calculations • Justification of answer with logic

Case Type: Acquiring a Company

Mergers & Acquisitions

General issue

Your client is considering acquiring a company and wants your opinion on the matter

Concepts often tested

- Ability to identify, organize, and articulate benefits and drawbacks (i.e., synergies)
- General knowledge of finance concepts
- General knowledge of competitive strategy

Key solution elements

- Cost-benefit analysis - short-term vs. long-term
- Rough valuation
- Identification of potential synergies or conflicts
- Expectations of competitive response

Case Type: Analyzing an Industry

Industry analysis		
General issue	Concepts often tested	Key solution elements
Your client wants to know about a particular industry and wants you to analyze it	• Ability to think broadly and completely • General business acumen • Ability to analyze within the context of a client problem	• Life-stage of industry • Analysis of competitors, suppliers and barriers • Competitive structure • Analysis of risks, substitutes, complements, potential innovations

Agenda

- ▶ Types of Cases

- ▶ Framing Strategy

- ▶ Common Reference Frameworks

- ▶ Applying Your Framework - Group Exercise

Step 1: Understand the Client's Goals

- ▶ Copy down the given information
- ▶ Identify the client's goals
- ▶ Verify your understanding by summarizing for the interviewer
- ▶ Ask for some time to come up with a plan

Step 2: Formulating a Plan

- ▶ Structure the problem
- ▶ If possible, form a tentative hypothesis
- ▶ Form your plan for approaching the problem:
 - For wide-open problems, plan to investigate the most relevant issues, then develop a hypothesis
 - For better-defined problems, plan to get information which proves your hypothesis

Step 3: Executing Your Plan

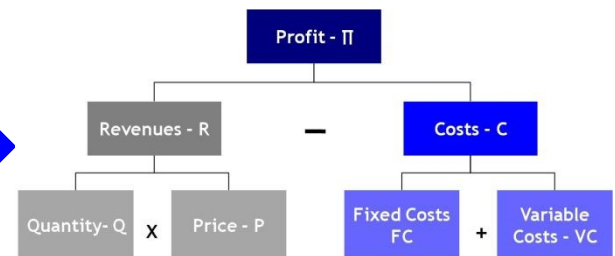
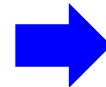
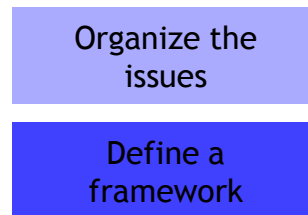
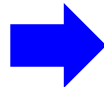
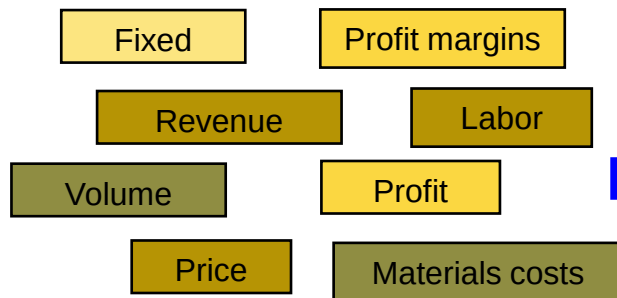
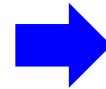
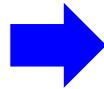
- ▶ Quickly describe your plan to the interviewer
- ▶ Begin the dynamic data-gathering process:
 - *Get the important information*
 - *Process that information*
 - *Update your hypothesis/plan*
 - *Repeat*
 - Move from the big-picture towards the details
- ▶ Deliver a quick summary and recommendation

Agenda

- ▶ Types of Cases
- ▶ Framing Strategy
- ▶ Common Reference Frameworks
- ▶ Applying Your Framework - Group Exercise

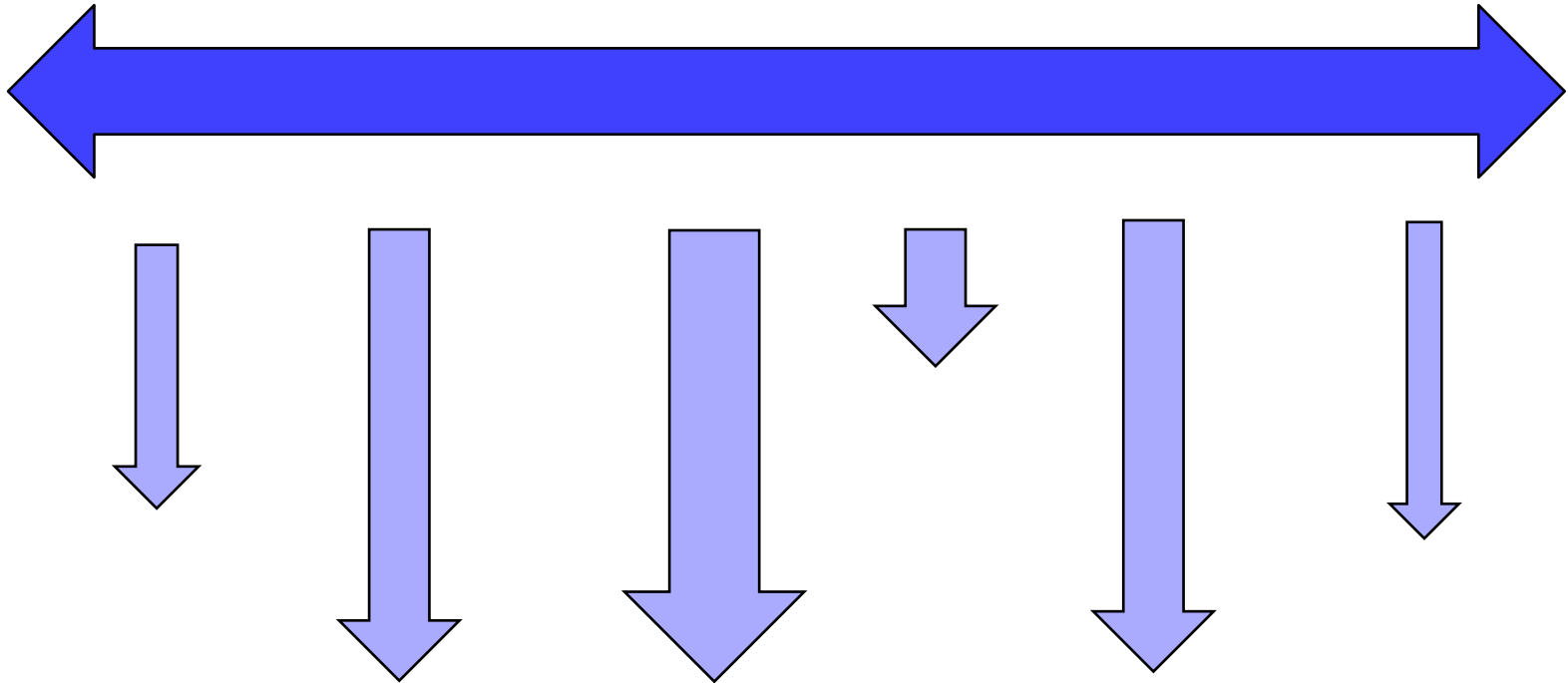
What is a framework and when to use it

- ▶ A framework is a structure that you use to organize your thoughts and analyze critical issues in a given case
- ▶ Frameworks= Road Map



MECE - Mutually Exclusive and Collectively Exhaustive

Break a problem into buckets and identify MECE sub-assertions from which to drill down further



The most common tools to construct a framework

MARKETING/STRATEGY CONCEPTS

- ▶ 4 C's
- ▶ 4 P's
- ▶ Contribution Analysis
- ▶ Sizing and Segmentation

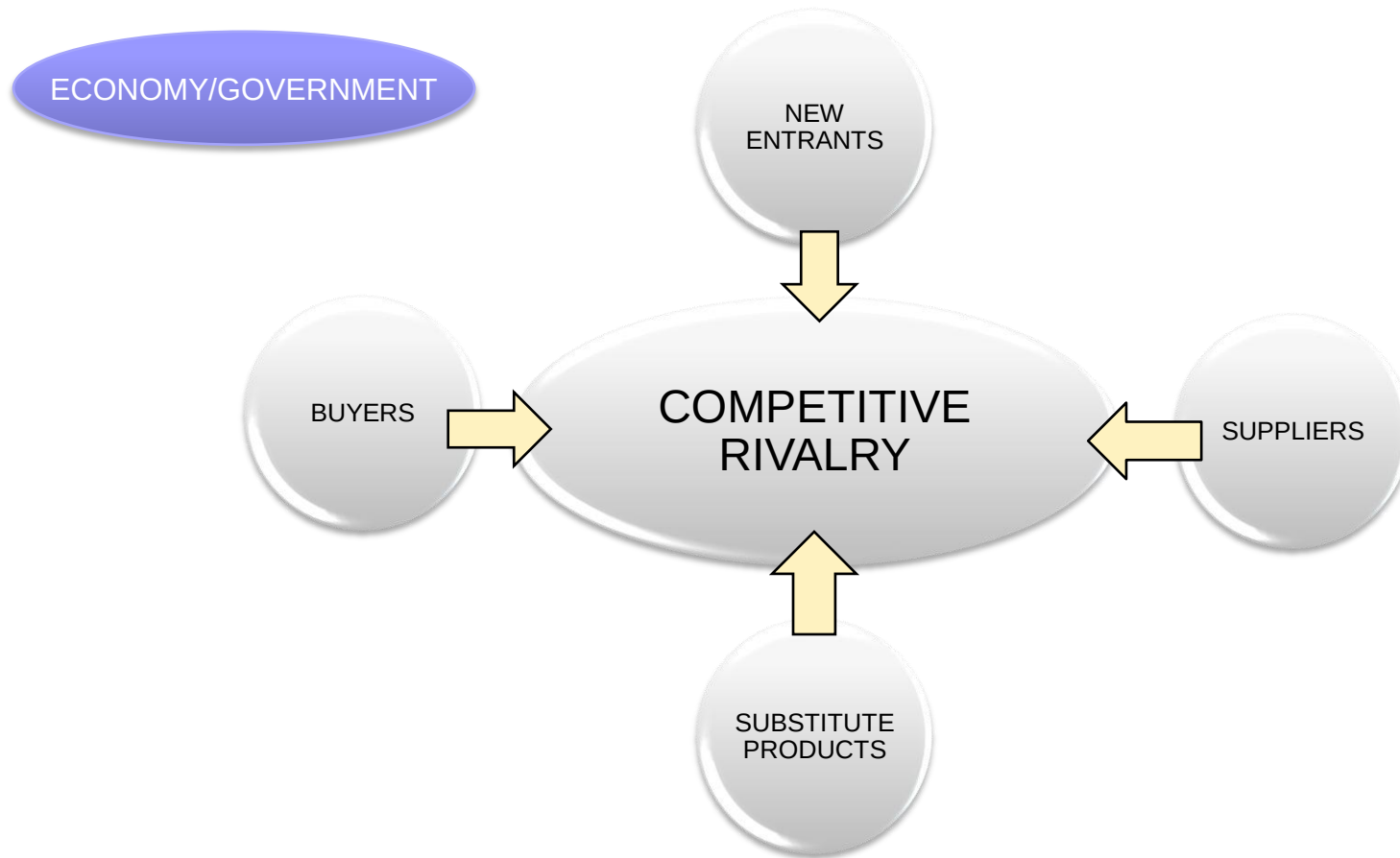
OTHER MAJOR CONCEPTS

- ▶ Porter's Five Forces
- ▶ SWOT
- ▶ Value Chain Analysis

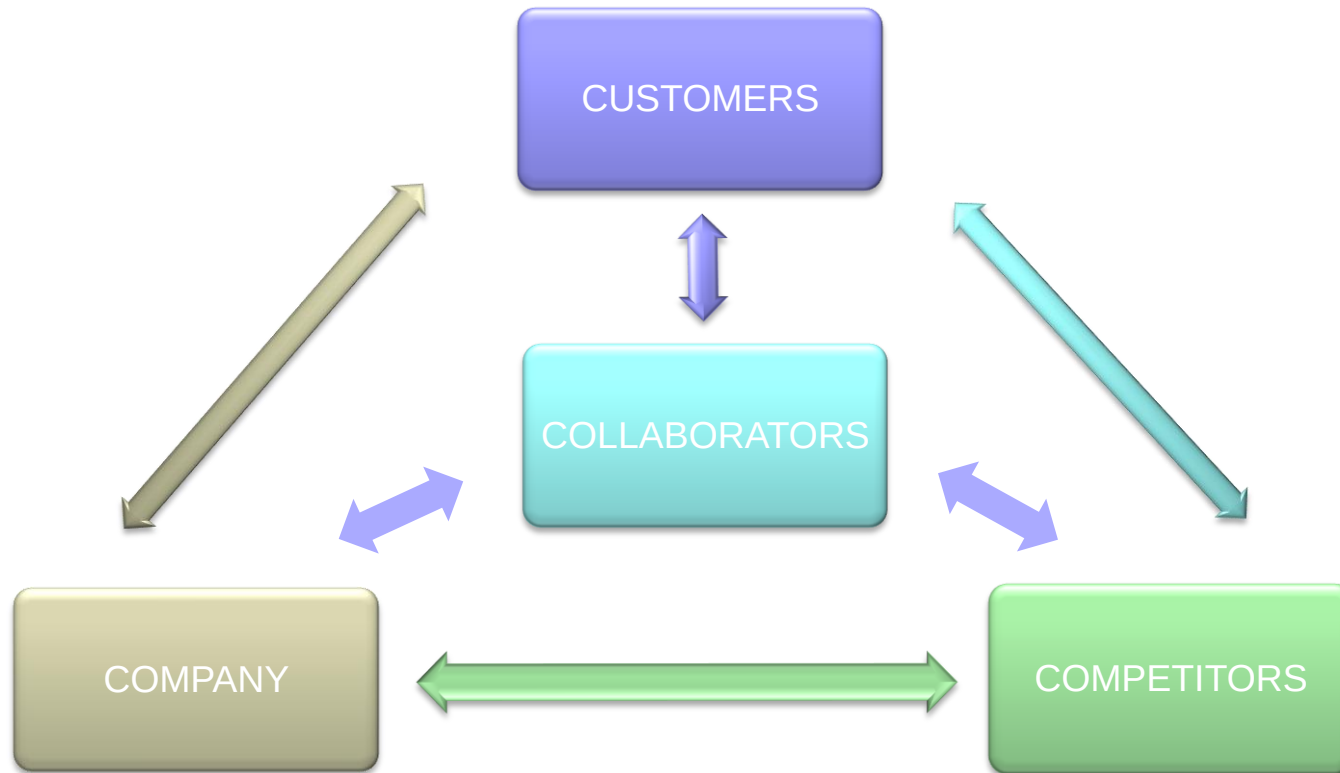
The most common tools to construct a framework

- ▶ Internal / External
- ▶ Cost / Revenues
- ▶ Breakeven Point
- ▶ Cost / Benefit
- ▶ Costs Analysis: Marginal, Fixed / Variable
- ▶ Supply / Demand
- ▶ Income Statement, Balance Sheet
- ▶ Net Present Value (NPV)
- ▶ CAPM and WACC

Porter's five forces

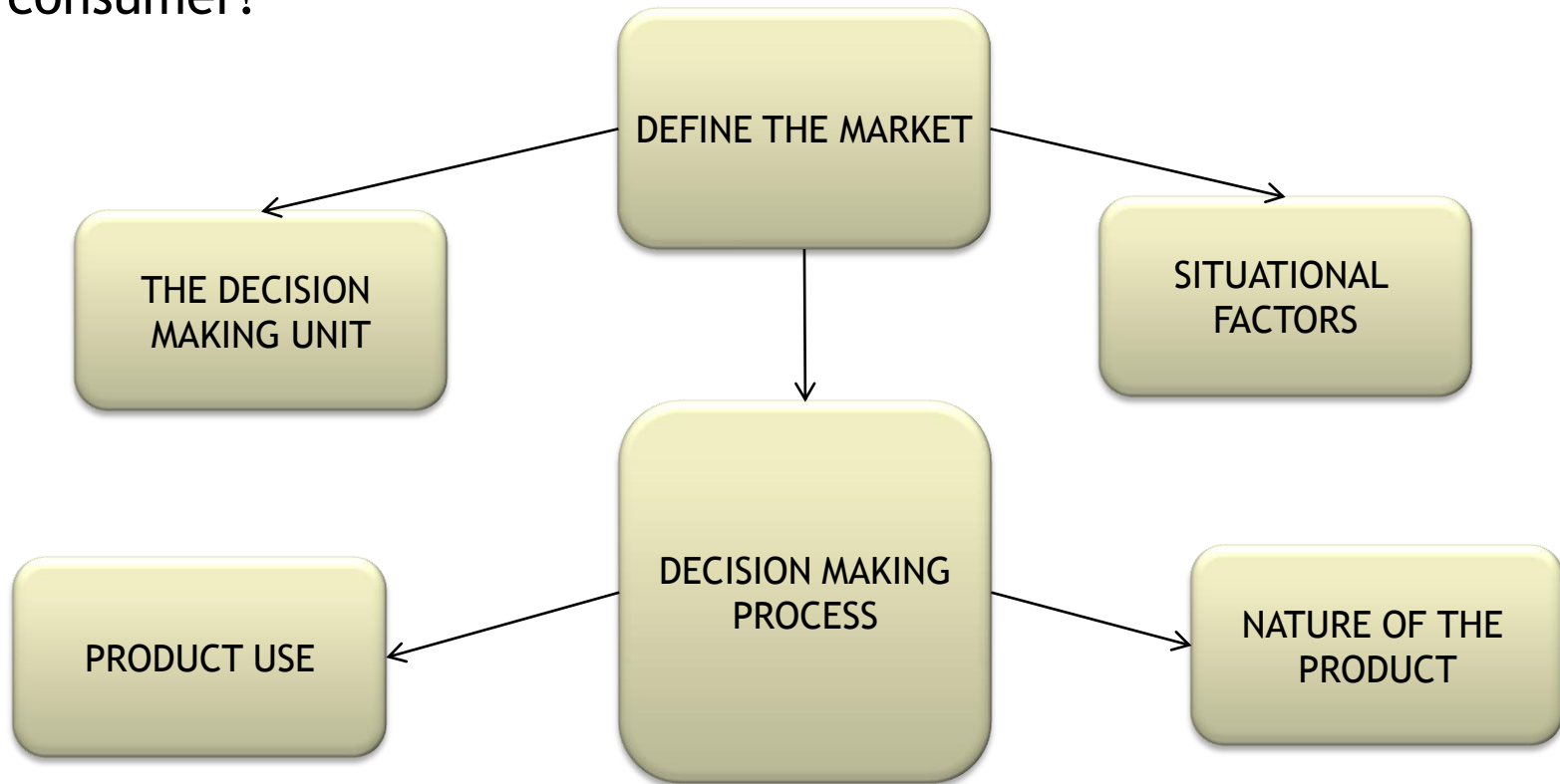


4 C's

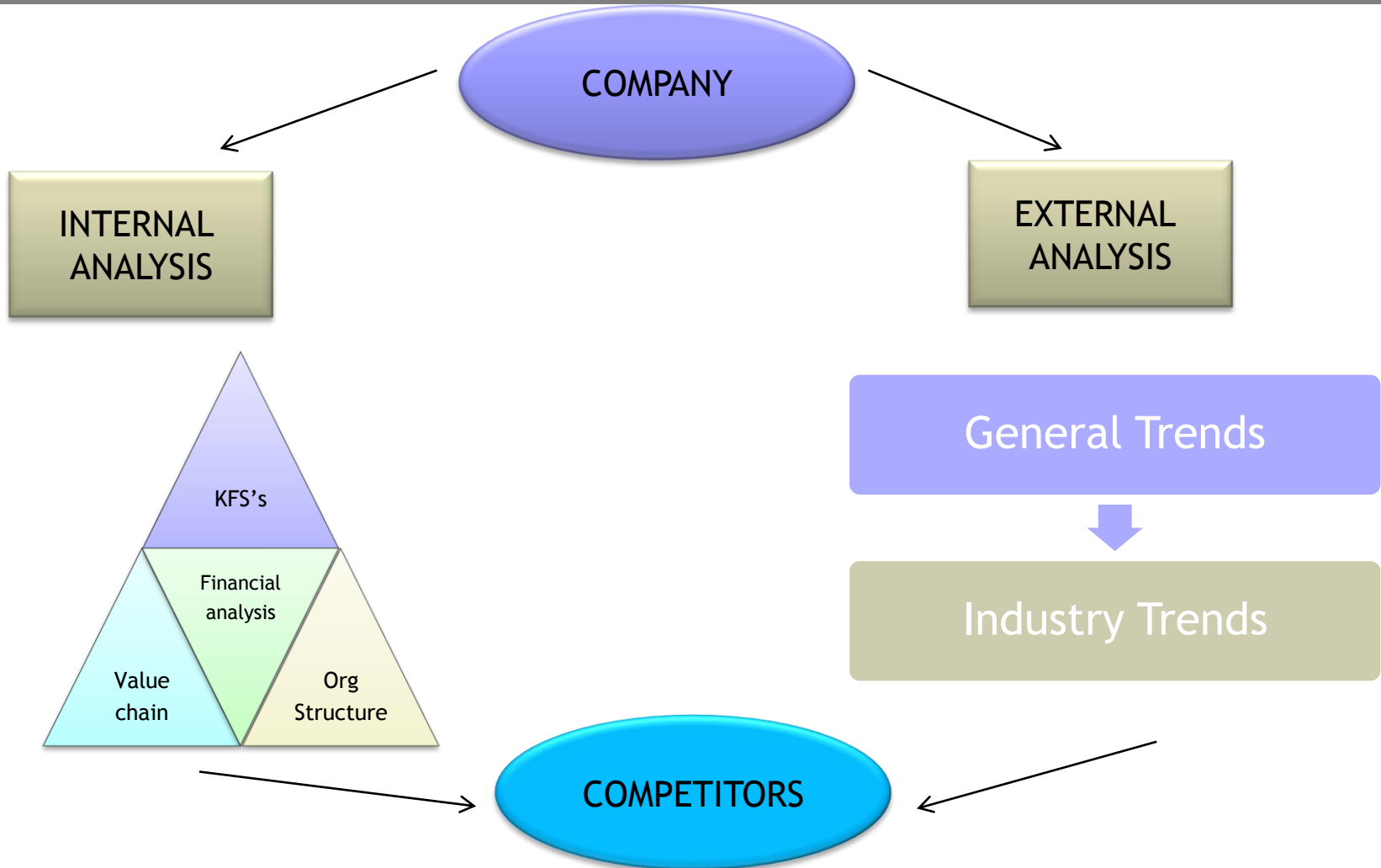


4 C's Customer

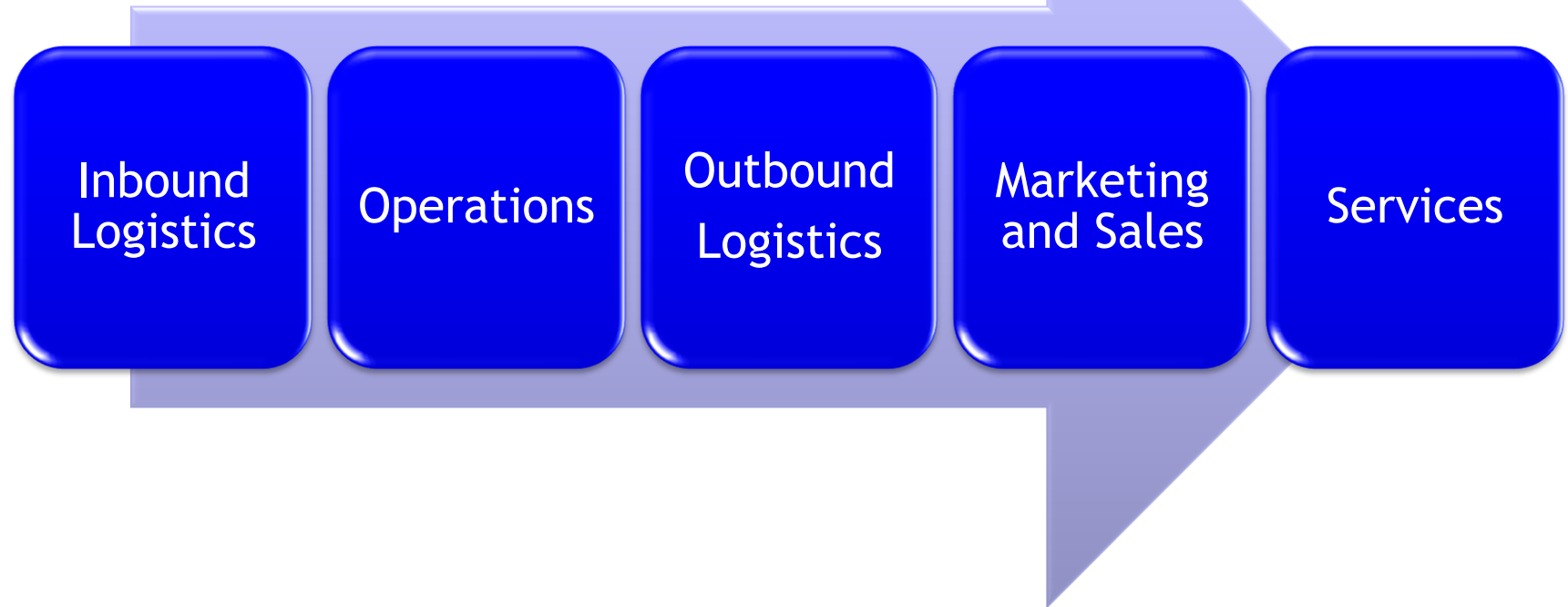
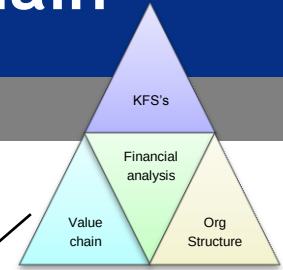
- ▶ Remember. Often your purchasing unit is different from your end consumer!



4 C's - Company



4 C's - Company - Internal Analysis - Value Chain

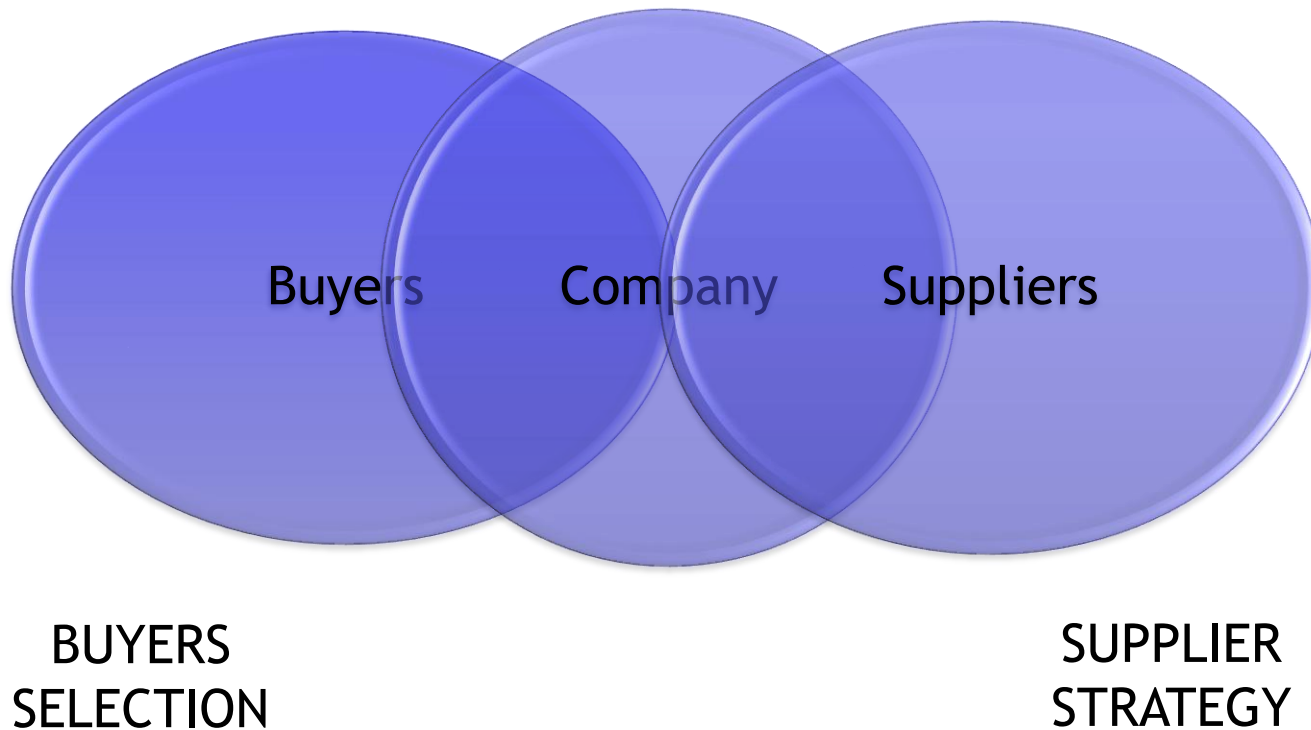


4 C's - Competitors

COMPETITORS

- ▶ The same company analysis previously discussed
- ▶ Other high level Concepts that can be used for Competitor's analysis
 - ❖ What drives the competitor: Goals and Assumptions
 - ❖ What is the competitor doing: Capabilities and Strategy

4 C's - Collaborators



QUESTIONS?

Agenda

- ▶ Common Reference Frameworks
- ▶ Types of Cases
- ▶ Framing Strategy
- ▶ Applying Your Framework - Group Exercise

Case Practice



Case: Rick's Kicks

Your client, Rick's Kicks, is a manufacturer of midrange and high-end athletic shoes. The company has been faltering in recent years and is under significant shareholder pressure to grow its business. A major mass merchant has expressed interest in launching an offering from your client in all of its stores. (The company currently does not sell to mass merchants.)

Should your company go ahead with the launch? If so, how?

Source: Ace Your Case (series), by WetFeet

Case: Car Manufacturer

Your client is a major car manufacturer with significant sales and brand equity. Though the company is doing well, the CEO is looking for incremental opportunities. A major area of concern is that customers' positive interactions with the brand are largely limited to the car-buying experience, which occurs on average every three years.

How would you increase customers' positive interactions with the brand?

Source: Ace Your Case (series), by WetFeet

Case: Minor League Baseball Team

The marketing VP of a minor league baseball team would like to attract more kids to attend games. He has proposed to the team's general manager that kids' tickets be discounted 50 percent for all upcoming season games. Your firm has served the team previously on an unrelated matter, and now the GM is calling you as a trusted advisor to get your point of view before he makes a decision on the kids' marketing plan.

What are some of the issues he should consider?

Source: Ace Your Case (series), by WetFeet

Case Practice - Tattoo Ink

- ▶ Tattoo ink is carbon-based pigment. However, a research lab in the Dominican Republic has developed a new erasable tattoo ink called DR Ink. This new patented technology is based on microencapsulated biodegradable and bio-absorbable dyes within safe, colorless polymer beads. This will allow a tattoo artist to ink his design on his customer that will last a life time, or if the ink is put under a special filtered light the tattoo will completely disappear.

DR Ink wants to know:

- ▶ What is the size of the European tattoo market?
- ▶ How many tattoos get inked every year?
- ▶ How should they enter this market?
- ▶ How should they price the ink?

Source: CQ interactive @ casequestions.com

Case Practice - Tattoo Ink

- ▶ In the European market, how many tattoos get inked every year?

Source: CQ interactive @ casequestions.com

Case Practice - Tattoo Ink

- ▶ What are the major ways that DR Ink can enter the market?

Source: CQ interactive @ casequestions.com

Case Practice - Tattoo Ink

- ▶ Do you think opening up their own tattoo parlor is a good idea? What do you think the market is like?

Source: CQ interactive @ casequestions.com

Case Practice - Tattoo Ink

- ▶ What are the pros and cons of distributing our own ink?

Source: CQ interactive @ casequestions.com

Case Practice - Tattoo Ink

- ▶ Assume that DR Ink has decided to manufacture the ink themselves. That the current ink market is made up of three players all with an equal share of market. Because our product is unique I'll assume that within 5 years we could have 40% of the market - if we price it right. How do we price the ink?
 - What are three ways to price a product?

Source: CQ interactive @ casequestions.com

Case Practice - Tattoo Ink

- ▶ Assume that carbon-based ink costs €4 to manufacture, ink manufacturers sell it to the distributor for €14 which is a 250% markup. Distributors sell the carbon-based inks for €20 for a 125 ml bottle.
- DR Ink can manufacturer their ink for €7 for a 125 ml bottle. Keeping in mind price-based costing at what price should DR Ink sell it's 125ml bottle to the distributors?

Source: CQ interactive @ casequestions.com



YALE SOM
CONSULTING CLUB

2008 Consulting Club Week 3 Commonly Used Academic Concepts

September 25, 2008

These written materials are not complete without accompanying verbal comments

Agenda

- ▶ Academic Concepts in Cases
- ▶ Competitive Strategy Concepts
- ▶ Economics Concepts
- ▶ Finance Concepts
- ▶ Best-Practices for Giving Cases
- ▶ Casing Exercises

Academic Concepts Show Up Frequently in Cases

Examples of academic concepts in case interviews

*“Your client is thinking about raising the price of their widget. What are some of the main issues they should consider before making this decision?
... If they raise the price by \$1, how much will their revenue increase or decrease?”*

“In the competitive structure of your client’s industry, we would say that they sit in the middle of the market. In recent years, they’ve been competing more fiercely with producers of high-end, premium widgets, and also with producers of low-end, generic widgets.”

“One of the issues with outsourcing widgets to China is that domestic (U.S.) demand is highly variable, and it takes 3-4 extra weeks to ship widgets from China to the U.S.”

Academic concepts and issues are generally embedded within the larger context of the case, though often they are the central issue of the case

Academic Concepts Show Up Frequently in Cases

Examples of academic concepts in case interviews

“Your client is thinking about raising the price of their widget. What are some of the main issues they should consider before making this decision? . . . If they raise the price by \$1, how much will their revenue increase or decrease?” (Elasticity - Basics of Economics)

“In the competitive structure of your client’s industry, we would say that they sit in the middle of the market. In recent years, they’ve been competing more fiercely with producers of high-end, premium widgets, and also with producers of low-end, generic widgets.”

(Trading Up/Trading Down - Competitor and Customer)

“One of the issues with outsourcing widgets to China is that domestic (U.S.) demand is highly variable, and it takes 3-4 extra weeks to ship widgets from China to the U.S.” (Inventory Management - Operations Engine)

Academic concepts and issues are generally embedded within the larger context of the case, though often they are the central issue of the case

Certain Concepts Often Show Up In Specific Industries, But Not Always

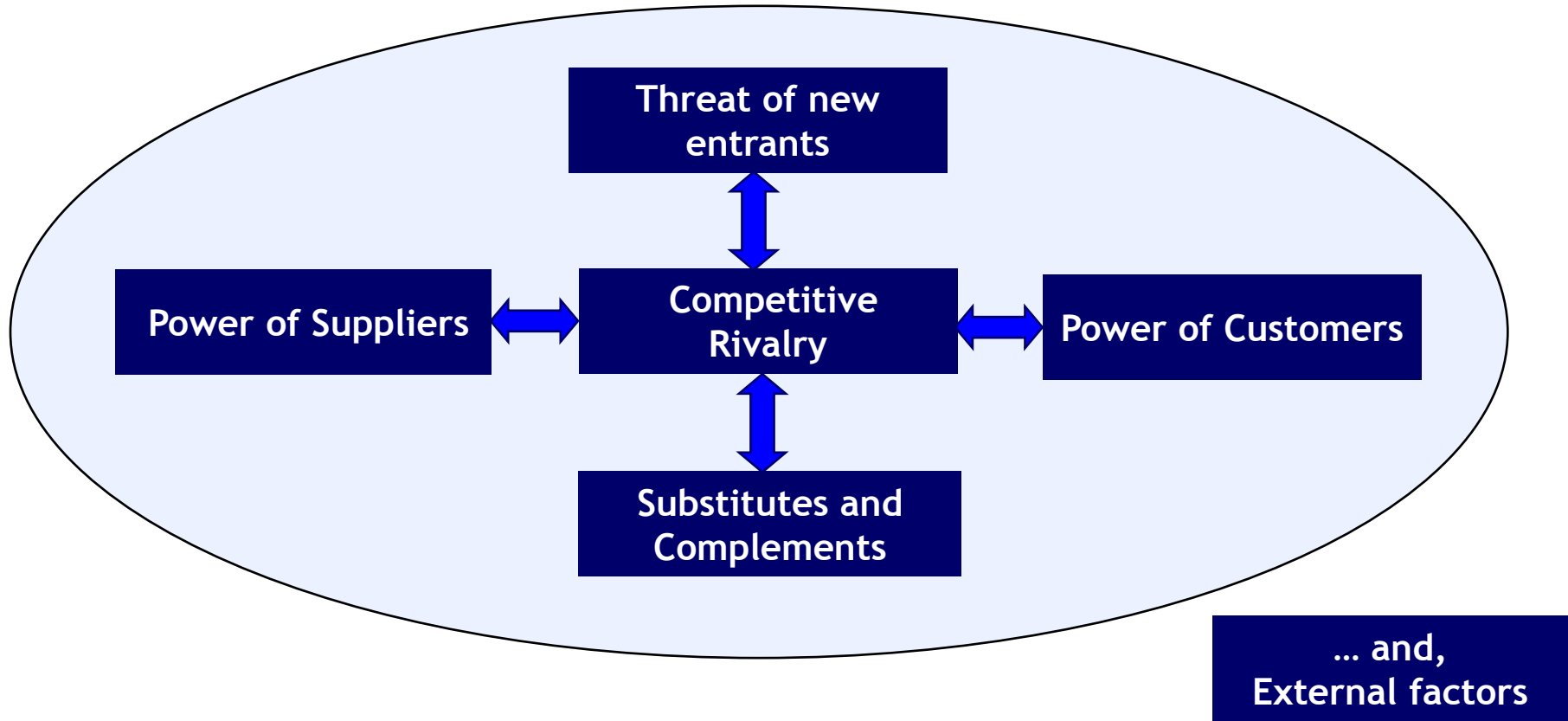
Industry	Important Concepts, Metrics, and Industry Characteristics
Hotel / Lodging	■ Occupancy rates and average daily room rate ■ Business travelers vs. vacationers
Airlines	■ Utilization rates ■ High fixed costs, high fuel costs ■ Business travelers vs. vacationers
Pharmaceuticals	■ High R&D costs, low marginal (variable) costs ■ Patents and monopoly pricing
Consumer Goods	■ Importance of segmenting customer base ■ Price sensitivity, 80/20 rule ■ Product mix
Trucking (Transportation & Logistics)	■ Utilization rates ■ Front- and back-hauling
Utilities	■ Monopolistic behavior ■ High fixed costs, costly to enter ■ Seasonal (daily?) demand fluctuations

Agenda

- ▶ Academic Concepts in Cases
- ▶ Competitive Strategy Concepts
- ▶ Economics Concepts
- ▶ Finance Concepts
- ▶ Best-Practices for Giving Cases
- ▶ Casing Exercises

Everyone Should Be Familiar With Industry Analysis

Porter's Five Forces



Industry Analysis Continued

	Ask about....	To understand
Competitive Dynamics	Market share of competitors Growth rate and profitability	Is the industry concentrated/fragmented Is there long-term potential
Buyer Power	Price sensitivity of buyers Loyalty of buyers	Do players have to compete on price Can you gain market share
Supplier Power	Competition amongst suppliers Access to raw materials	Can you can lower costs through negotiation Will procurement will be expensive
New entrants	Barriers to entry (patents, capital requirements, etc)	Can you enter the industry and can you prevent others from following you in
Substitutes/ Complements	Products that can replace your product and products that go well with your product	Are there products that can reduce (or enhance) your product's demand
External factors	Changes in buyer demographics State of the economy	Is the target segment growing/shrinking Can buyers afford your product

Industry Analysis is Often Used as the First Step in a Market Entry Case

Steps in a Market Entry Analysis:

▶ Size of the market opportunity:

- Market sizing

▶ Profitability:

- Costs: What will our fixed and variable costs be?
- Price: What will be our selling price? (How much are consumers willing to pay?)
- Revenues: How much of the market do we hope to capture? When can we expect to break even?

$$\begin{aligned}\text{Profit} &= \text{Revenues} - \text{Costs} \\ &= (\text{Market size} * \text{Market share}) - \text{Costs}\end{aligned}$$

▶ Possible added benefits from market entry:

- Diversification
- Synergies
- Competitive Advantage
- Growth

After Assessing the Attractiveness of Market Entry, the Next Step is to Answer the “How” Questions

The “How” of Markey Entry

Mode of entry

- Independent - “organic” growth
- Acquire a competitor in the market
- Establish a joint venture

Capital requirements

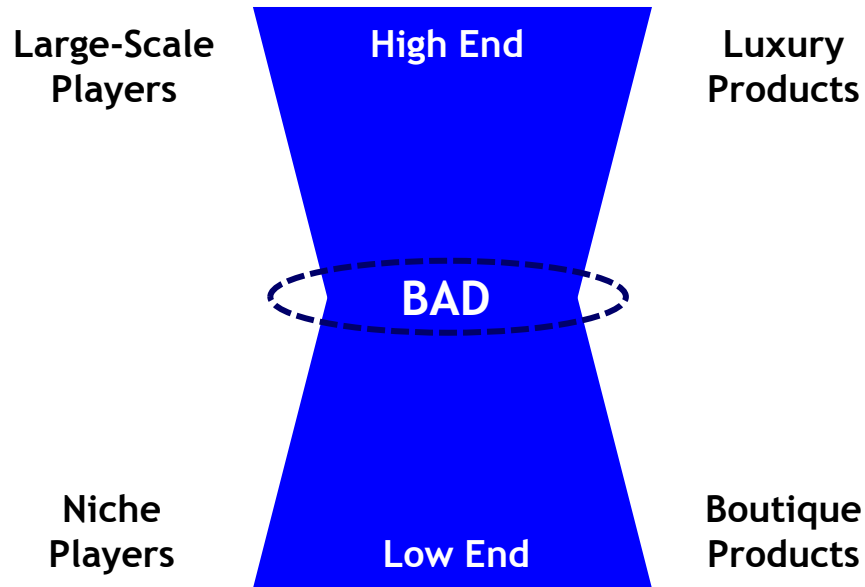
- Financial capital
 - Debt vs. equity
 - Impacts of influence from investors
- Human capital
 - Competencies
 - Technical expertise
 - Experience

Exit strategy

- Divest
- Spin-off
- Timing of exit
- Risks

Industry Positioning - where to be in the market

Companies Often Get Squeezed in the Middle . . .



. . . Which Leads to Trading Up / Trading Down

Agenda

- ▶ Academic Concepts in Cases
- ▶ Competitive Strategy Concepts
- ▶ Economics Concepts
- ▶ Finance Concepts
- ▶ Best-Practices for Giving Cases
- ▶ Casing Exercises

Sample Basic Economic Concepts

Concept	Application & Comments
Price Elasticity	- Understand how to calculate, not just what it means - Understand inelastic goods vs. elastic goods - Affected by brand loyalty, availability of substitutes, long-term vs. short-term
Substitutes & Complements	- Can have major effect on market share (application to market share cases) - Often an important issue in consumer goods cases
Scale Economies (MES)	- Extremely important in manufacturing cases, industries with factory issues - Understand declining average cost - How can you deal with excess capacity?
Monopoly	- Certain industries prone to monopoly - pharma, regulated utilities, etc. - Understand monopoly pricing, barriers to entry, predatory pricing
Marginal Costs	- Firms do not produce in an industry if $P < MC$!!!! - Understand why having lower MC than competitors is an advantage
Basic Supply & Demand	- The most basic principle, but very important - Review what shifts supply & demand mean - why might a market be in decline?

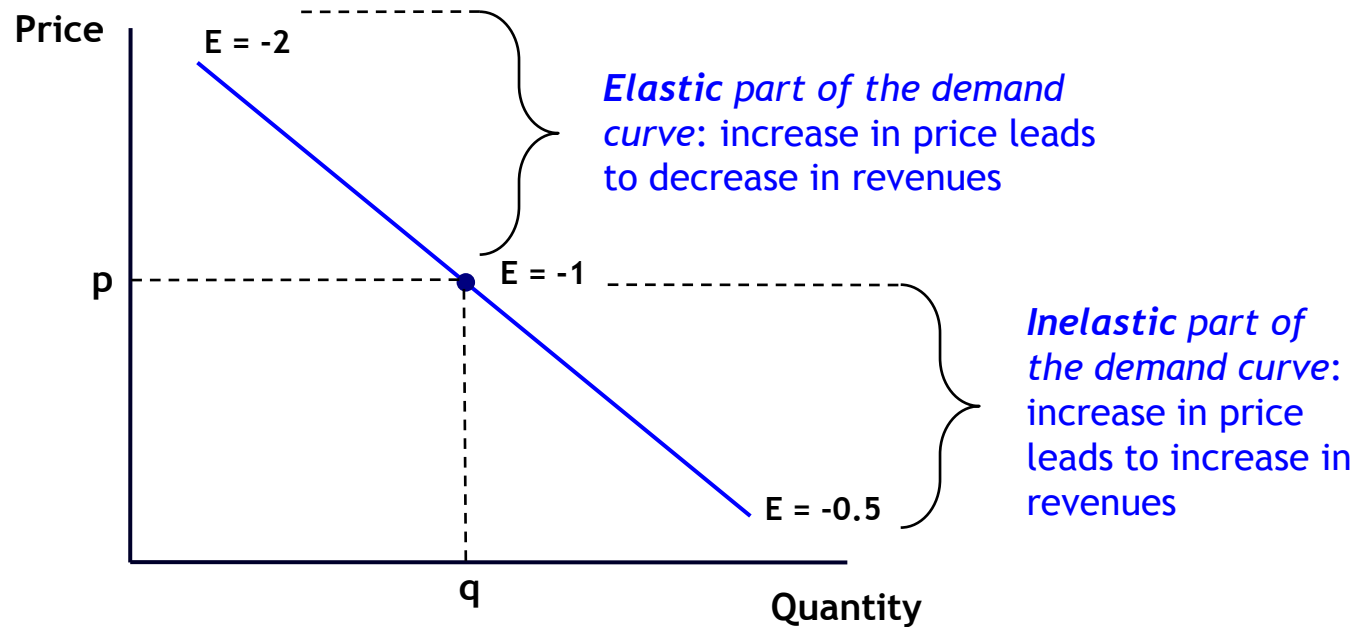
Elasticity

Elasticity:

$$E = \frac{\% \Delta Q}{\% \Delta P}$$

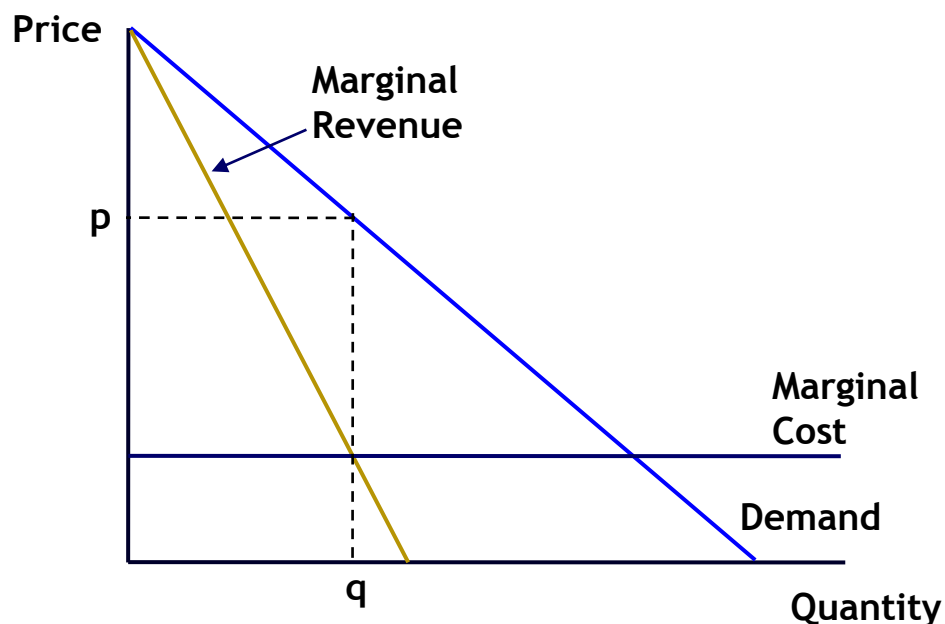
$$E = \frac{P}{Q} \times \frac{1}{\text{Slope}}$$

Demand Curve Elasticity



Monopolies

Monopoly Price & Quantity



In a monopoly, price is set for the demand curve where marginal revenue equals marginal cost

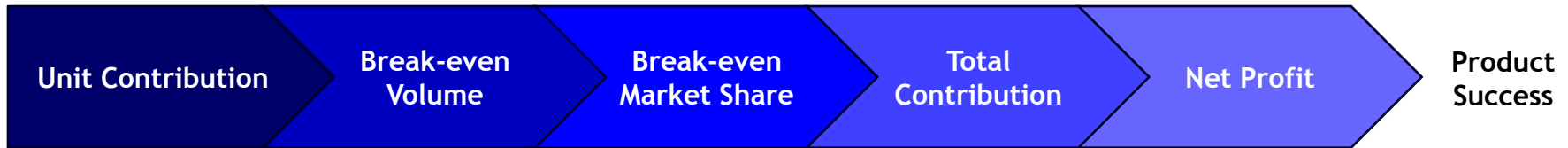
Sources of Monopoly

- High fixed costs, and significant economies of scale
- High R&D costs (intellectual property)
- High marketing expenses
- Network effects
- Government ownership/regulation

Monopolies cannot simply charge as much as they want, because there are almost always substitutes, and at some price consumers will become elastic

Contribution Analysis

Contribution analysis can help you understand the economics of the contribution of a product. This framework can be used to evaluate or forecast the overall success of a product



Tools

Examples

Calculate the contribution that each unit provides to cover fixed/overhead costs

Unit Contribution

$$\begin{array}{r} \text{Unit Selling Price} \\ - \text{Variable Cost} \\ \hline = \text{Unit Contribution} \end{array}$$

$$\begin{array}{r} + \$50.00 \\ - \$21.25 \\ \hline \$28.75 \end{array}$$

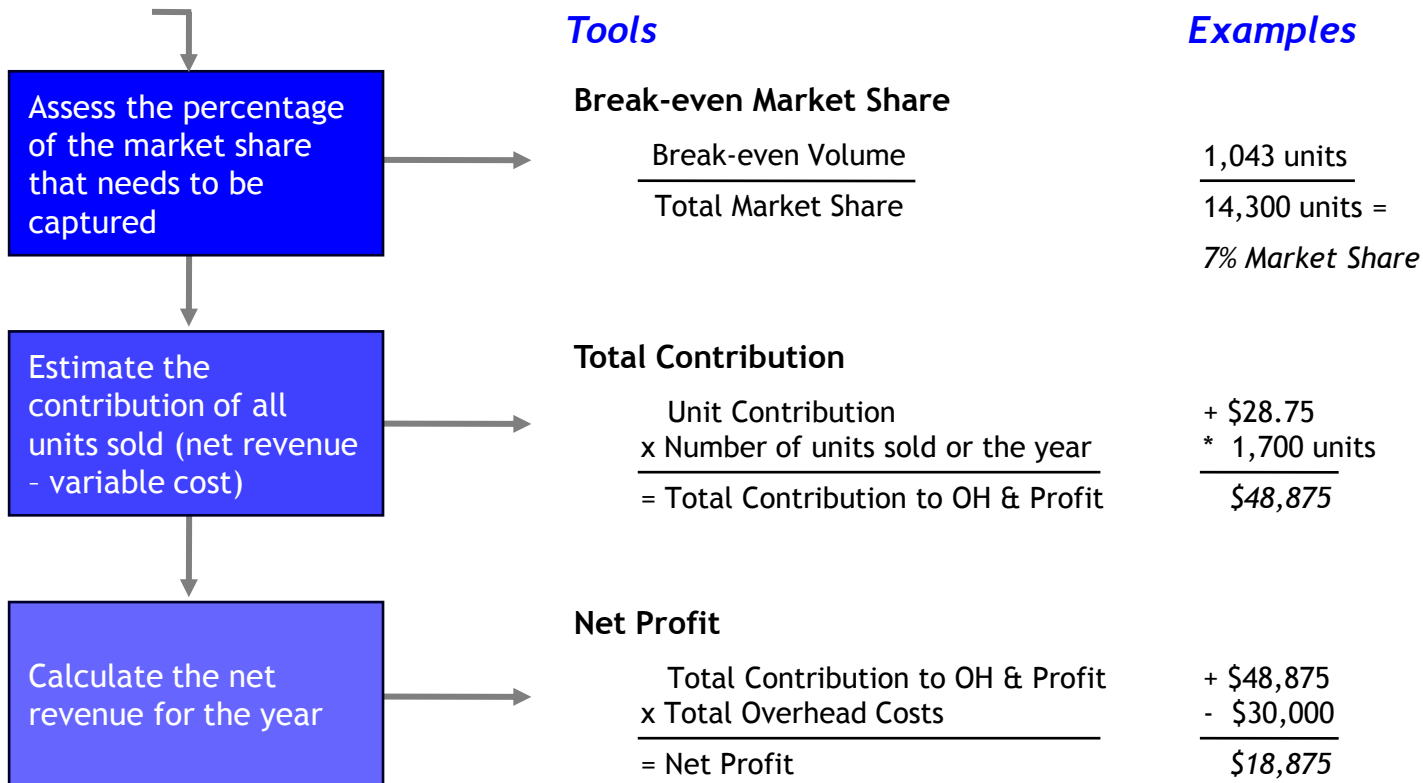
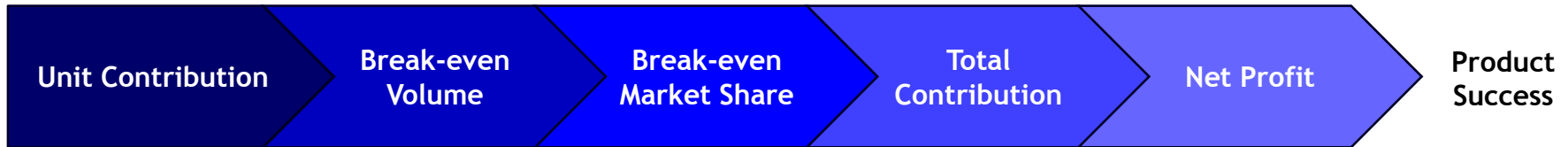
Determine the number of unites that need to be sold to break even

Break-even Volume

$$\begin{array}{r} \text{Fixed Costs} \\ \hline \text{Unit Contribution} \end{array}$$

$$\begin{array}{r} \$30,000 \\ \hline \$28.75/\text{unit} = \\ 1,043 \text{ units} \end{array}$$

Contribution Analysis (cont.)



Market Sizing

Determine the estimated number of customers in your market segment

Start

Number of customers targeted

X

Number of purchases made per period

X

Number of units per purchase

X

Price per unit

Think about how this differs when you consider different groups within your market segment

What price are the different segments/consumer groups willing to pay?

=

Total revenue in market

Finish

Market Share

Start

Total population in question

X

Number of customers targeted

X

Percent share of product type

X

The company's percent share of product type

=

Company's % share of the market

Finish

Can you increase the demand for the product in your targeted area?

Can you expand the market for a particular product type? (e.g., mountain bike vs. speed bike)

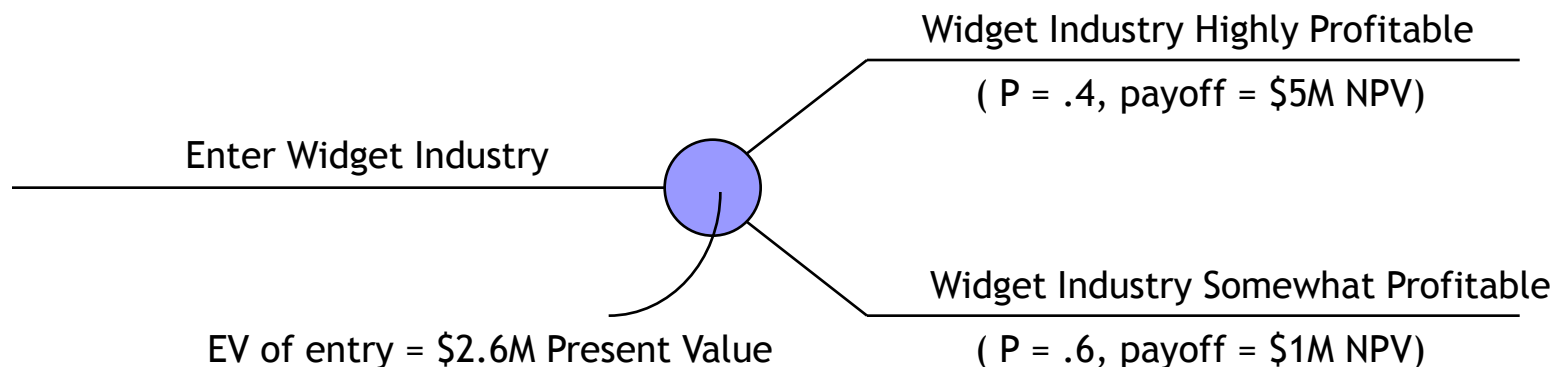
Can the company increase its product type?

Agenda

- ▶ Academic Concepts in Cases
- ▶ Competitive Strategy Concepts
- ▶ Economics Concepts
- ▶ Finance Concepts
- ▶ Best-Practices for Giving Cases
- ▶ Casing Exercises

Sometimes an Expected Value Calculation is Needed

Expected Value and Net Present Value



Simple Net Present Value Calculation:

$$\begin{array}{rcl} \$2.6\text{M} & \rightarrow & \text{EV of entry - initial investment cost} \\ - \$1.5\text{M} & \rightarrow & \text{(Fixed) initial investment cost} = \$1.5\text{M} \\ \hline \text{NPV } \$1.1\text{M} & \rightarrow & \text{Decision: Enter widget industry} \end{array}$$

Key idea - be able to recognize when doing an expected value calculation is called for

Other Important Finance Concepts

(We will visit these in more detail later in the semester)

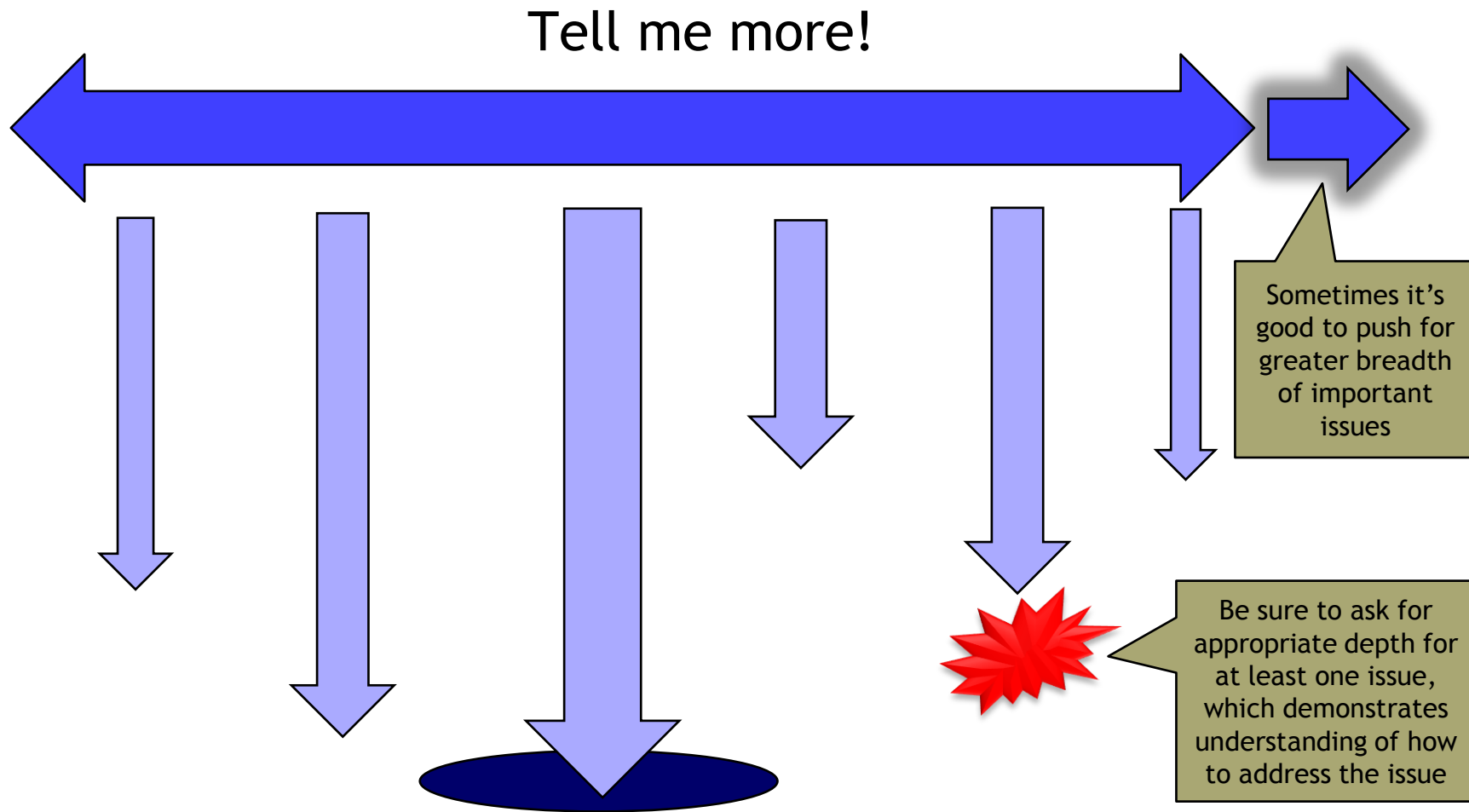
- Net present value calculations
- Cost accounting (semi-variable costs)
- Cost of capital
- Stock price . . . discounted future cash flows . . .
but what else do investors care about?

**Finance knowledge less important than in banking interviews . . .
but more important to consulting than you might think**

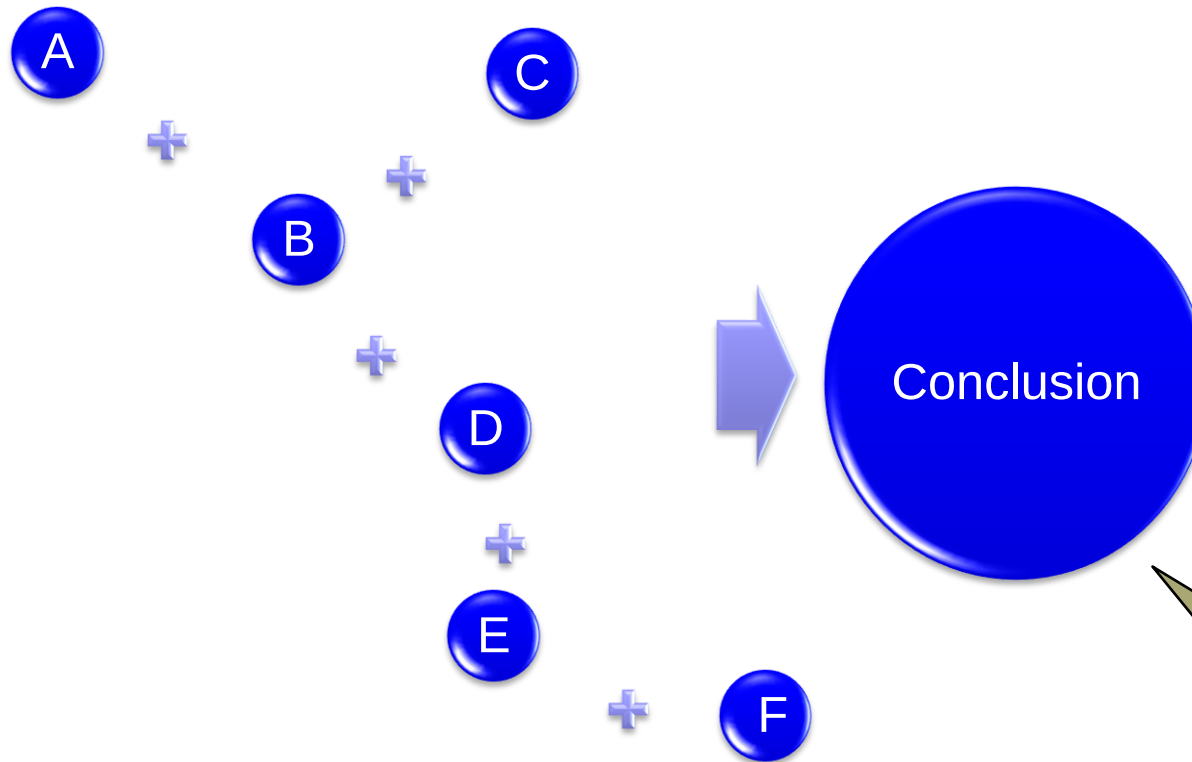
Agenda

- ▶ Academic Concepts in Cases
- ▶ Competitive Strategy Concepts
- ▶ Economics Concepts
- ▶ Finance Concepts
- ▶ Best-Practices for Giving Cases
- ▶ Casing Exercises

Ask for Breadth and Depth

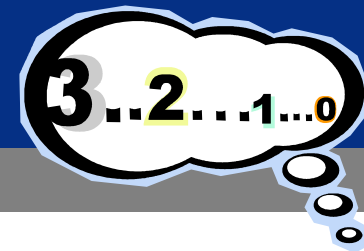


Coach on Integration of Ideas for a Strong Conclusion



Be sure to encourage your case-taker to state a well-synthesized, concise conclusion. Ask your case-giver to repeat the conclusion several times if necessary

Always do the NUMBERS!!!



Know Formulas



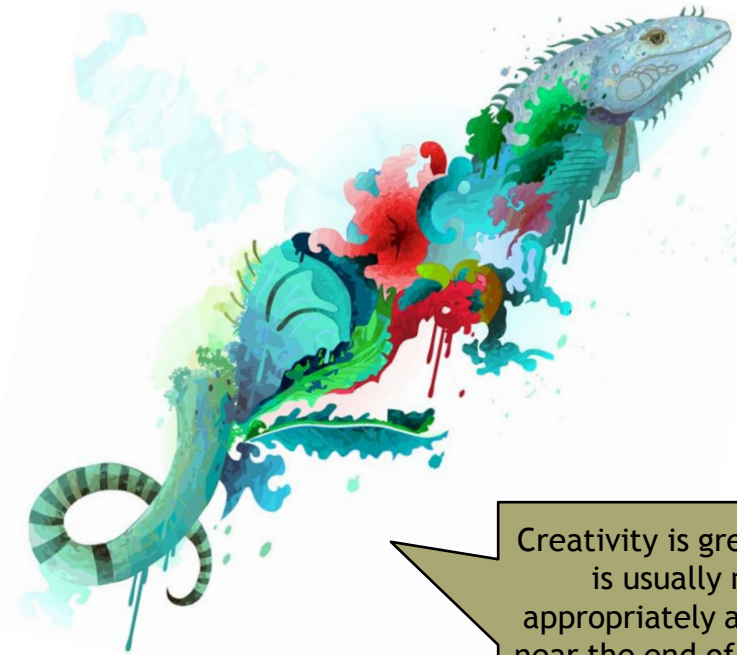
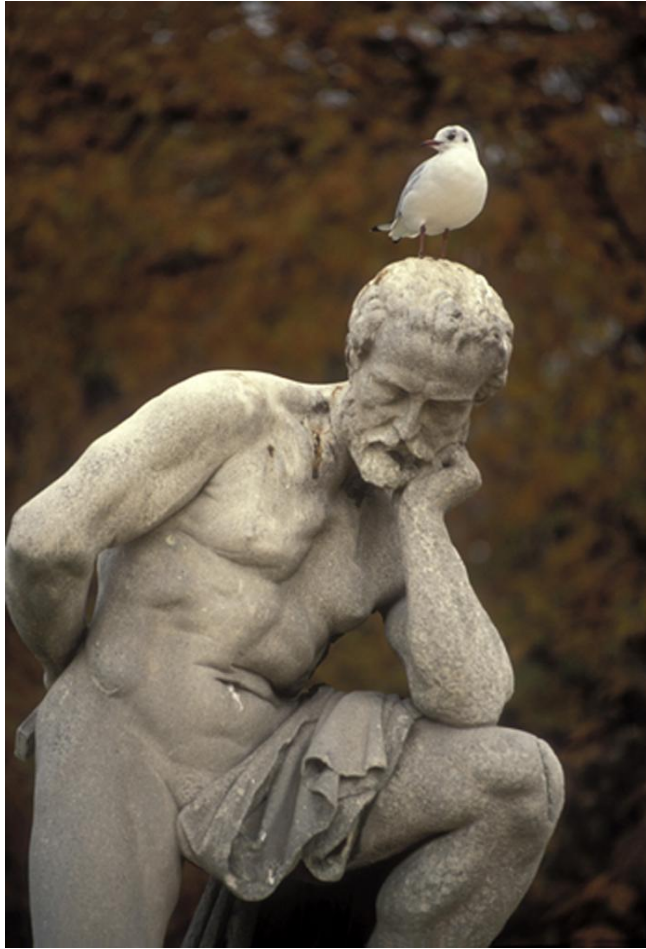
Bring Your
Numbers
Back to
Profits and
Costs



See Trends



Encourage Creativity - Where it Fits With the Context of the Case



Creativity is great! But it is usually most appropriately addressed near the end of the case. “Nudge” your case-taker to address these items later in the case if necessary

Agenda

- ▶ Academic Concepts in Cases
- ▶ Competitive Strategy Concepts
- ▶ Economics Concepts
- ▶ Finance Concepts
- ▶ Best-Practices for Giving Cases
- ▶ Casing Exercises

Case Practice - Percent Change

- ▶ The Copperline Picayune is a regional weekly newspaper covering 5 counties. It has seen its overall circulation drop from 220,000 to 210,000 this last year. Its ad revenues have dropped 10% this last year alone. Overall revenues were down 10%. On-line ad revenues were up 15%, archive retrieval fees were up 20%. If it is lucky, The Picayune will breakeven this year, but the owners aren't feeling lucky.
- What was the percentage drop in circulation?

Source: CQ interactive @ casequestions.com

Case Practice - Percent Change

- ▶ What was the percentage drop in circulation?

Suggested Answer

- ▶ $220,000 - 210,000 = 10,000$
 $10,000 / 220,000 = 0.045$
- ▶ 4.5%

Source: CQ interactive @ casequestions.com

Case Practice - Percent Change

- ▶ US Box Office revenues for 2007 increased 5.4% to \$9.6 billion. What were Box Office Revenues in 2006?

Source: CQ interactive @ casequestions.com

Case Practice - Percent Change

Suggested Answer

$$X * 1.054 = 9.6$$

$$X = 9.6 / 1.054$$

$$X = 9.11$$

\$9.11 billion

Source: CQ interactive @ casequestions.com

Case Practice - Breakeven Points

- ▶ DCS makes grinders for large sail boats. They spent \$720,000 to retool the plant. They have a unit cost of \$40. Their distribution, labor and marketing costs 20% of the sale price. The sale price is \$100. Assuming that costs remain the same and that all units sell at \$100, what is their breakeven point in units sold?

Source: CQ interactive @ casequestions.com

Case Practice - Breakeven Points

► **Suggested Answer**

- ▶ Revenue per unit \$100
 - COGS per unit (40)
 - Distribution, etc cost per unit (20)
-
- ▶ Profit per unit \$40
-
- ▶ Divide the profit per unit into the retooling costs - $720,000 / 40 = 18,000$

Alternatively

$$20x + 40x + 720,000 = 100x$$

$$60x + 720,000 = 100x$$

$$720,000 = 40x$$

$$x = 18,000$$

Source: CQ interactive @ casequestions.com

Case Practice - Net Present Value

- ▶ Hector wants to give back to his community. He asked the principal of his middle school to identify 5 promising 7th grade students. Hector funded college accounts for the students. Each account received \$10,000. In five years when the students have graduated from high school, and if they have been accepted to a college, they can withdraw the money. If the interest rate was 5% how much will each account be worth?

Source: CQ interactive @ casequestions.com

Case Practice - Net Present Value

Suggested Answer

$$FV = 10,000 (1+.05)^5$$

$$FV = 10,000 (1.05)^5$$

$$FV = 10,000 (1.2762)$$

$$FV = 12,762.00$$

\$12,762

Source: CQ interactive @ casequestions.com

Case Practice - Breakeven Points

- ▶ You have developed a web-site that trains law students for the bar exams. It cost you \$15,000 to develop, \$1,200 a year to maintain and you spent \$10,000 to market it. Students pay \$50 for six months of access. How many individual customers do you need to breakeven?

Source: CQ interactive @ casequestions.com

Case Practice - Breakeven Points

► Suggested Answer

► Costs $\$15,000 + \$1200 + \$10,000 = \$26,200 / 50$

► = 524 students

Source: CQ interactive @ casequestions.com

Case Practice - Breakeven Points

- ▶ Our client wants to open up private security checkpoints at the airport. Instead of standing and waiting in a security line for 10-20 minutes with everyone else, passengers can pay \$5 a head and zip through in less than a minute. You need to figure out the monthly breakeven point — how many passengers. Here is all the info.

Fixed Monthly Costs

Airport rent – \$2,000

X-ray rental – \$5,000 for 2 machines

Variable Costs

Labor - \$15 an hour (includes benefits)

Hours of operation: 7am to 5pm – 5 days a week (M-F) – assume 4 weeks per month

Number of staff needed at all times is 5.

Source: CQ interactive @ casequestions.com

Case Practice - Breakeven Points

Suggested Answer

- ▶ Monthly rent of airport space and x-ray machines is \$7,000
Labor - 10 hours a day, 5 days a week - 50 hours a week times 5 people equals 250 hours a week or 1000 hour a month * \$15/hour = 15,000
- ▶ Our total costs are \$22,000 a month.
 $22,000 / \$5 = 4,400$ passengers
- ▶ Breakeven is 4,400 a month or 220 passengers a week day

Source: CQ interactive @ casequestions.com

Case Practice - Net Present Value

- ▶ A book publisher is currently negotiating a royalty agreement with one of its authors. The standard agreement calls for an annual \$200,000 payment to the author for each of the next three years. The author, however, wants to buy a new home for his lovely bride, and doesn't believe in mortgages. He is proposing a one-time payment of \$500,000 from the publisher. If the publisher can afford such a large cash payment, should the publisher take the deal? He checks with the bank and current interest rates are 5 percent.

Source: CQ interactive @ casequestions.com

Case Practice - Net Present Value

Suggested Answer

- ▶ Year = 1 $(200,000) / 1.05^1 = 190,476$
Year = 2 $(200,000) / 1.05^2 = 181,406$
Year = 3 $(200,000) / 1.05^3 = \underline{172,768}$
- ▶ **Present value** \$544,650
- ▶ Yes, the publisher should pay the lump sum.

Source: CQ interactive @ casequestions.com

Case Practice - Frameworks

- ▶ American Bank, a \$450 billion asset bank in the US, is thinking of expanding its reach in the Northeastern US. As such, it is considering purchasing a local bank with a large footprint in the New England area.
- What are the areas you would look at when considering a particular acquisition target?

Source: CQ interactive @ casequestions.com

Case Practice - Frameworks

- ▶ What are the areas you would look at when considering a particular acquisition target?

Industry/ Market Analysis:

- ▶ What is the size and growth of the local New England banking market?
- ▶ How do companies compete?
- ▶ Are there any other large national banks or is it based on local companies only?
- ▶ Are there any government / other barriers to entry?

Analyze the Target:

- ▶ Benchmark with competitors — in performance, share, and profit
Look at strengths/weaknesses within the local market

Synergies:

- ▶ Look at potential synergies:
 - Cost — administrative, increased efficiencies, economies of scale
 - Revenue — cross-selling opportunities of existing products.

Potential Issues:

- ▶ Are the cultures compatible/ different? Is technology a big barrier to merger?

Financing:

- ▶ How do we plan on financing the acquisition?

Source: CQ interactive @ casequestions.com

Case Practice - Pricing

- ▶ GE, they bring good things to life, has invested a new light-bulb that never burns out. It could burn for more than 500 years and it would never blink. The director of marketing calls you into her office and asks, “How would you price this?”
- ▶ What do you tell her?

Source: CQ interactive @ casequestions.com

Case Practice - Pricing

- ▶ GE, they bring good things to life, has invested a new light-bulb that never burns out. It could burn for more than 500 years and it would never blink. The director of marketing calls you into her office and asks, “How would you price this to the consumer?”
- ▶ What do you tell her?

Source: CQ interactive @ casequestions.com

Case Practice - Pricing

- ▶ GE, they bring good things to life, has invested a new light-bulb that never burns out. It could burn for more than 500 years and it would never blink. The director of marketing calls you into her office and asks, “How would you price this to the consumer?”
- ▶ What facts do we need to answer this question?
 - Industry related questions: Patents, other similar products, etc.
 - Development Costs: It cost \$20mm to develop this product (sunk cost)
 - Production Costs of a traditional light bulb: \$.05 to manufacture, sold to distributor then to store then to the consumer for \$.75
 - Production Costs for new bulb: \$5.00 to manufacture
 - Risks?

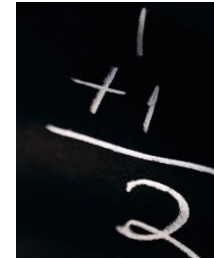
Source: CQ interactive @ casequestions.com



YALE SOM
CONSULTING CLUB

2008 Consulting Club Week 4

Math !



October 2, 2008

These written materials are not complete without accompanying verbal comments

Educating Leaders for Business and Society



YALE SCHOOL OF MANAGEMENT

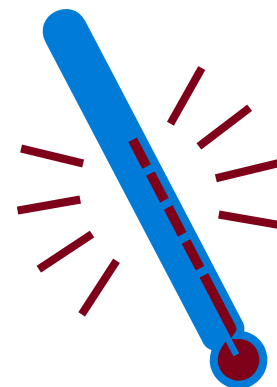
Agenda

- ▶ Why Do They Try to Trick Us with Math?
- ▶ Fundamental Math Concepts
- ▶ Intermediate Math Concepts
- ▶ Advanced Math Concepts
- ▶ Group Casing Exercises

Why numbers matter in the case interview

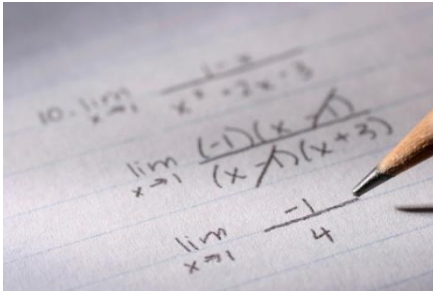
“I know that numbers mean everything in business. But why are math skills important in interviews if I’ll have a calculator at work?”

- Basic skills demonstrate that you can think methodically and understand mathematical relationships
- Firms want to see if you can apply numerical data to clarify a business story and substantiate a set of recommendations
- Interviews force you to articulate the logic of your calculations
- Tedious math problems test your poise under pressure



So You Don't Like Math.

How often do the more complex math concepts really show up in cases?



Answer: Complex math shows up less often than basic division, percents, etc. BUT, you never know when you will get a case with tricky math

How should you prepare?

- Practice basic math - multiplication, division, decimals, percentages
- Get familiar with the basic equations for a short list of more complex concepts
- Develop an *understanding* of the purpose of more complex math concepts, and know the general impact of answers of different magnitude

Agenda

- ▶ Why Do They Try to Trick Us with Math?

- ▶ Fundamental Math Concepts

- ▶ Intermediate Math Concepts

- ▶ Advanced Math Concepts

- ▶ Group Casing Exercises

Fundamental Math Techniques

- ▶ You **MUST** know these and practice them:
 - Long division
 - Multiplication
 - Addition
 - Subtraction
 - Basic fractions and simplification
 - Percentages
 - Algebra, solving for X

Large Numbers

► Difficulty:

- Easy to make mistakes (especially number of zeroes)
- Time-consuming to write (e.g. \$25,000,000)

► Solution:

- Symbol of magnitude: \$25M
- Scientific notation: $\$2.5 \times 10^7$
- $1\text{K} = 1.0 \times 10^3 \dots 1\text{M} = 1.0 \times 10^6 \dots 1\text{B} = 1.0 \times 10^9$
- Working with scientific notation:
 - Multiplying: Multiply coefficients and add exponents.
 - Adding: Exponents must be equal!

Percentages

- ▶ Difficulty:
 - Takes different forms, each using similar language
- ▶ Solution:
 - Part/whole concept ($\%$ = part/whole)
 - Understand “percent of” and “percent increase”
 - Understand markup and margin
- ▶ Tips:
 - Practice!
 - Purplemath.com covers all the bases

Percentages: you thought they were easy?

► Question:

- Your company sells ice cream in a market where customers are segmented by favorite flavor. 70% of your customers belong to the vanilla segment. Your customers make up 3.6% of the market. 10% of the customers in the market prefer vanilla ice cream. What percentage of the vanilla customers in the market does your company serve?

Percentages: you thought they were easy?

► Question:

- Your company sells ice cream in a market where customers are segmented by favorite flavor. 70% of your customers belong to the vanilla segment. Your customers make up 3.6% of the market. 10% of the customers in the market prefer vanilla ice cream. What percentage of the vanilla customers in the market does your company serve?

► Answer:

- $0.7 * 0.036 / 0.1 = 0.252$
We are introduced to three overlapping groups of people: “your customers,” “vanilla customers,” and “the market.” We are asked how a particular overlap - the intersection of all three groups - compares to one of the groups - the vanilla customers. Note: everything is in percent terms relative to the market.
- **Step 1:** find the “part”: 70% of 3.6% = 2.52%. **Step 2:** find the “whole”: we already know it’s 10%. **Step 3:** divide part/whole: $2.52\% / 10\% = 25.2\%$

Agenda

- ▶ Why Do They Try to Trick Us with Math?
- ▶ Fundamental Math Concepts
- ▶ Intermediate Math Concepts
- ▶ Advanced Math Concepts
- ▶ Group Casing Exercises

Market Sizing - Segmentation

▶ Difficulty:

- How do you size a market quickly and sensibly?

▶ Solution:

- Start with a known population
- Divide the population into segments according to [age, income, etc]
- Pick percentages of each segment who belongs in your market

▶ Tips:

- Use 300M for US Population
- Use manageable round numbers for bracket sizes and percentages
 - $300/4 = 75 \dots 75/3 = 25$

Market Sizing - Segmentation Example

► Question:

- How many cups of coffee are sold in the US every year?

► Answer:

- Total cups/day * days/year
- The US Pop is 300M. Let's bracket by age in groups of 20-year spans and assume an even distribution (75M in each bracket).

age	pop	% drink coffee	subtotal	cups/day	total
0-20	75M	1/10	7.5M	1	7.5M
20-40	75M	2/3	50M	2	100M
40-60	75M	1/3	25M	2	50M
60+	75M	1/3	25M	1	25M
					182.5M

$$182.5M * 365 = 66.6B \text{ cups/yr}$$

Market Sizing

Determine the estimated number of customers in your market segment

Start

Number of customers targeted

X

Number of purchases made per period

X

Number of units per purchase

X

Price per unit

Think about how this differs when you consider different groups within your market segment

What price are the different segments/consumer groups willing to pay?

=

Total revenue in market

Finish

Rule of 72

Rule of 72 helps to answer the following questions

- ▶ How long it takes to double your savings (approximately) for a given saving rate

Start with the number 72.

Divide by the rate of return you expect to earn

You will get your investment horizon, or number of years you need to double your investments

- ▶ What rate of return should you get if you want to double your money in a given number of years

Start with 72.

Divide by the number of years you want to have your return doubled

You will get the AFTER-TAX COMPOUND ANNUAL RATE OF RETURN you would have to earn

Question

- ▶ In investor knows she can earn 16% of return on her money. How long it will take to double the investment?
- ▶ In investor needs to double his money in 3 years (legally). What is the rate of return must he earn to do this successfully?

Solution

- ▶ In investor knows she can earn 16% of return on her money. How long it will take to double the investment?

$$72/16 = 4.5 \text{ years}$$

- ▶ In investor needs to double his money in 3 years (legally). What is the rate of return must he earn to do this successfully?

$$72/3 = 24\%$$

Estimation

► What's 14% of 2.3 million?

Solution

■ Estimation

- *Look for 15% of 2.3MM*
- *10% of 2.3MM = 230K*
- *50% of 230K = 115K*
- *Answer ~ 345K*
- *(You could then easily reduce your estimate to 325K by subtracting 20K.)*

■ Calculation

- *Scale down and scale up*
- *10% of 2.3MM = 230K*
- *1% of 2.3MM = 23K*
- *4% of 2.3MM = 92K*
- *230K + 92K = 322K*

Zeros and Decimals

► What's .0003% of 2.6 billion?

Solution

■ *Estimate/Calculation*

- *Where does the decimal go?*
- *Find .0001% and scale up.*
- *Dividing .0001 by 100 gives a % multiplier of:*

. 0 0 0 0 0 1
~~~~~

- *From R to L, the decimal point has moved 6 places.*
- *Moving the decimal point left 6 places in 2.6 billion = 2,600*
- *2,600 x 3 = 7,800*



# Percentage

► What's 14% of 2.3 million?

# Percentage Solution

- What's 14% of 2.3 million?
- Estimate
  - Look for 15% of 2.3MM
  - 10% of 2.3MM = 230K
  - 50% of 230K = 115K
  - Answer ~ 345K
  - (You could then easily reduce your estimate to 325K by subtracting 20K.)
- Calculation
  - Scale down and scale up
  - 10% of 2.3MM = 230K
  - 1% of 2.3MM = 23K
  - 4% of 2.3MM = 92K
  - 230K + 92K = 322K

# Problem

In 2030 the world will have 1.2 billion cars. This is one vehicle for every 6.7 people projected to be on the planet at 2030. This is also 92% more cars than we have on the road today.

- ▶ How many vehicles are on the road today?
- ▶ If the ratio of cars to people will be half of the one we have today, how many people are in the world today?

# Solution

**Solve:** 625 million cars

$$X * 1.92 = 1,200,000,000$$

**Estimate:**  $1,200,000,000 / 2 = 600$  million , You are dividing by a little less than 2, so it will be a little more than 600. Estimate solution is 630,000,000.

**Solve:** 6.979 Billion

$$X * 1.92 = 6.7 * 2$$

$$X = 13.4 / 1.92 = 6.9$$

**Estimate:** From  $(X * 1.92 = 6.7 * 2)$  you should see that the number is a little higher than 6.7. Estimate solution is 7 billion.

(Btw, current world population is 6.7 billion)

# Gross Margin

## CONCEPT

- Gross margin tells us how much of every dollar earned by the firm, after paying direct fixed and variable costs, is available to cover overhead costs and serve as a buffer to unknown items
- More simply, GM tells us how many of our sales dollars are profit

## Gross Margin Equation

$$GM = \frac{(\text{Revenue} - \text{Costs})}{\text{Revenue}}$$

## Components of the GM Equation

$$\text{Gross Profit} = \text{Revenues} - (\text{FC} + \text{VC})$$

$$GM = \frac{\text{Gross Profit}}{\text{Revenue}}$$

► Revenue = 3.5MM. Cost = 2.1MM. Find GM.

# Solution

$$GM = (R - C) / R$$

- Estimate
  - *1.5 / 3.5 is a little under 50%.*
  - *Answer ~ 45%*
  
- Calculation
  - $(3.5 - 2.1) / 3.5$
  - $1.4 / 3.5$
  - *Answer = 40%*

# Agenda

- ▶ Why Do They Try to Trick Us with Math?
- ▶ Fundamental Math Concepts
- ▶ Intermediate Math Concepts
- ▶ Advanced Math Concepts
- ▶ Group Casing Exercises



# Net Present Value

## CONCEPT

- Net present value tells us whether a particular investment will add value to our company. If NPV is positive, make the investment
- Literally, NPV is a comparison of the expected future cash inflows from an investment to the expected cash outlay(s) needed to finance the project

### Step 1: PV of Cash Inflows

$$\text{Present Value} = \frac{\text{Cash Flow}}{(1 + r)^t}$$

### Step 2: Find NPV

$$\frac{\text{PV of Cash Inflows} - \text{Initial Investment}}{= \text{NPV}}$$

\* Initial Investment represents the PV of all cash outflows needed to finance the project

# Example of Present Value Discounting

## Common discount rates and periods that show up in cases

|               |       | Number of periods |      |      |      |      |
|---------------|-------|-------------------|------|------|------|------|
| Discount rate | \$100 | 1                 | 2    | 3    | 4    | 5    |
|               | 3%    | 97.1              | 94.3 | 91.5 | 88.8 | 86.3 |
|               | 5%    | 95.2              | 90.7 | 86.4 | 82.3 | 78.4 |
|               | 7%    | 93.5              | 87.3 | 81.6 | 76.3 | 71.3 |
|               | 10%   | 90.9              | 82.6 | 75.1 | 68.3 | 62.1 |
|               | 12%   | 89.3              | 79.7 | 71.2 | 63.6 | 56.7 |

## Remember:

- The larger the discount rate, the smaller the present value
- The greater the number of periods, the smaller the present value
- The larger the discount rate, the greater the risk premium applied to the potential investment
- Use the *Rule of 72* when possible
- Round and estimate when possible

# Present Value of an Annuity

## CONCEPT

- The present value of an annuity, also known as the perpetual growth formula, tells us the value today of a series of cash flows that expect to last forever into the future
- The perpetual growth formula can help us value a company

## Basic Equation

$$\text{Present Value} = \frac{\text{Coupon}}{r - g}$$

## Definition of Terms

*Coupon* = the regular cash flow payment

$r$  = discount rate

$g$  = growth rate

# Customer Lifetime Value

## CONCEPT

- Customer Lifetime Value is the present value of all future streams of profits that a customer generates over the life of his/her business with the firm
- CLV tells us how much we can spend to “acquire” a new customer

## Basic Equation

$$CLV = m \left[ \frac{r}{1 + i - r} \right]$$

## Definition of Terms

$m$  = margin from each customer

$r$  = retention rate

$i$  = discount rate

# ACCOUNTING RATIOS

## Why it is important to know the key ratios:

- ▶ Used to evaluate the overall financial condition of a corporation
- ▶ Financial ratios allow for comparisons
  - between companies (when firms use the same accounting methods)
  - between industries
  - between different time periods for one company
  - between a single company and its industry average

# What do you want to remember

- SOLVENCY
- LIQUIDITY
- PROFITABILITY
- OPERATING EFFICIENCY
- FINANCIAL LEVERAGE

Values used in calculating financial ratios are taken mostly from the

**BALANCE SHEET, INCOME STATEMENT, CASH FLOW STATEMENT!**

# When to use in a case

- ▶ When you are given financial statements to analyze
- ▶ When you are trying to decide whether a company should invest in a new project now (a liquidity issue) or will it stay in business or be able to take on some extra debt (solvency issue)

# Solvency Measurements

## WHY?

To be able to assess the ability of a corporation to meet its long-term fixed expenses and to accomplish long-term expansion and growth.

- ▶ A ratio used to measure a company's ability to meet long-term obligations. The solvency ratio measures the size of a company's after-tax income, excluding non-cash depreciation expenses, as compared to the firm's total debt obligations. It provides a measurement of how likely a company will be to continue meeting its debt obligations.

$$\text{Solvency Ratio} = \frac{\text{After Tax Net Profit} + \text{Depreciation}}{\text{Long Term Liabilities} + \text{Short Term Liabilities}}$$



# Liquidity Measurements

$$\text{Current Ratio} = \frac{\text{Current Assets}}{\text{Current Liabilities}}$$

- ▶ Current Ratio answers the question whether the firm can meet its short term obligations.

$$\text{Quick Ratio} = \frac{\text{Cash \& Equivalents} + \text{Short - term Investments} + \text{Accounts Receivable}}{\text{Current Liabilities}}$$

- ▶ Quick Ratio answers the question whether a firm can meet short term obligations tomorrow.

# Problem

► At December 31 a company's records show the following information:

- Cash \$ 10,000
- Accounts Receivable 30,000
- Inventory 80,000
- Prepaid Insurance 6,000
- Long-term Assets 200,000
- Accounts Payable 30,000
- Note Payable due in 10 months 25,000
- Wages Payable 5,000
- Long-term Liabilities 70,000
- Stockholders' (Owner's) Equity 196,000

**What are the Current Ratio and the Quick Ratio?**

# Solution

- ▶ Current Assets - \$126,000:
  - Cash \$ 10,000
  - Accounts Receivable 30,000
  - Inventory 80,000
  - Prepaid Insurance 6,000
- ▶ Current Liabilities
  - Accounts Payable 30,000
  - Note Payable due in 10 months 25,000
  - Wages Payable 5,000
- ▶  $126,000 / 60,000 = 2.1$

# Liquidity measurements cont.

$$\text{Accounts Receivable Turnover} = \frac{\text{Net Credit Sales}}{\text{Average Accounts Receivable}}$$

- ▶ The larger the ratio, the shorter the collection period is for receivables, meaning that the company gets cash more quickly

Account receivable collection period =  $365 \text{ days} / \text{accounts receivable turnover ratio}$

- ▶ Reflects the number of days on average it takes the company to collect accounts receivables

# Profitability Measurements

## Why?

Give an idea of how likely it is that a company will turn a profit, as well as how that profit relates to other important information about the company.

$$\text{Net Profit Margin} = \frac{\text{Net Income}}{\text{Net Sales (Revenue)}}$$

- ▶ Decreasing profit margins year over year - changing market conditions, increasing competition, rising costs. Also, if a company's profit margin is out of line compared to the rest of its industry, it is worth the extra effort to find out why.

# Profitability Measurements cont.

$$\text{Operating Profit Margin} = \frac{\text{Operating Profit}}{\text{Net Sales (Revenue)}}$$

- ▶ Operating margin is a measurement of what proportion of a company's revenue is left over after paying for variable costs of production such as wages, raw materials, etc.
- ▶ look at the change in operating margin over time and to compare the company's yearly or quarterly figures to those of its competitors. If a company's margin is increasing, it is earning more per dollar of sales. The higher the margin, the better.

For example, if a company has an operating margin of 12%, this means that it makes \$0.12 (before interest and taxes) for every dollar of sales.

# Operating Efficiency Measurements

WHY?

Asset Turnover Ratio = Net Sales / Average Total Assets

- ▶ The higher is the ratio, the more efficiently a firm is using its assets

Rate of Return on Investments (known as ROI) = Net Income / Total Average Assets

ROI = Profit Margin \* Asset Turnover Ratio

# Financial Leverage

WHY?

To increase return on Equity

Financial Leverage =  $\frac{\text{Average Total Assets}}{\text{Average Common Stockholder's Equity}}$

- ▶ Amount of assets financed by stockholders



# Rate of Return

Rate of return on common stockholder 's equity

=

Net Income/Average Common Stockholder's Equity

=

Profit margin\*Asset turnover\*Leverage

# Agenda

- ▶ Why Do They Try to Trick Us with Math?
- ▶ Fundamental Math Concepts
- ▶ Intermediate Math Concepts
- ▶ Advanced Math Concepts
- ▶ Group Casing Exercises



**YALE SOM**  
**CONSULTING CLUB**

# **2008 Consulting Club Week 5**

October 9, 2008

These written materials are not complete without accompanying verbal comments

*Educating Leaders for Business and Society*



**YALE SCHOOL OF MANAGEMENT**

# Agenda

- ▶ Club Administrative Details
- ▶ Case Performance Discussion
- ▶ Beyond the Case
- ▶ Telling your story with the SOAR framework
- ▶ Case Practice

# Agenda

- ▶ Club Administrative Details

- ▶ Case Performance Discussion

- ▶ Beyond the Case

- ▶ Telling your story with the SOAR framework

- ▶ Case Practice

# Interview Overview - Case Performance Model

1. Case interview is also an interview of interpersonal skills
2. When the case is presented, make sure you fully understand the question and write it down
3. Take a minute to capture your thoughts on paper
4. Briefly walk your interviewer through your framework. Explain the path you want to take, outlining your rationale for choosing it
5. Ask relevant questions to gain further insight
6. Do not rush to get to any solution

# Interview Overview - Case Performance Model

7. Not all cases have "a right answer". Some approaches that are better than others. Stay focused on the problem at hand
8. Where possible, use easy numbers whenever you are estimating market size, price, costs, etc.
9. Develop clear and decisive recommendations

# Interview Overview - Summarizing the Case

- ▶ The best conclusions are concise and consistent with the facts given and assumptions that you have made
- ▶ Take a stand, back up your reasoning with logic and facts
- ▶ Begin with your recommendation
- ▶ Don't be tempted to hedge in your conclusion. Cases generally ask for a recommendation:
  - “Should company X invest in expanding production?”
  - “What is the market for Product X?”



# Interview Overview - Summarizing the Case

Bad: “We looked at a number of factors in evaluating this problem and results show that expanding production is good because it will lead to increased market share and will help defend against the strengthening competition. On the other hand, it may increase fixed costs to the point that the firm becomes unprofitable.”

Better: “Company X should expand its production. Doing so is the best strategy to increase market share and defend against competition. In order to maximize profitability, Company X will also need to place strict oversight and control over fixed costs during the expansion.”

# Agenda

- ▶ Club Administrative Details
- ▶ Case Performance Discussion
- ▶ Beyond the Case
- ▶ Telling your story with the SOAR framework
- ▶ Case Practice

# Beyond the Case: Behavioral Issues

- ▶ Behavioral /fit interview questions
  - Series of standardized questions designed to get you to talk about how you handled or responded to certain situations in the past
- ▶ Each answer will describe situations from your past and the feelings and observations about them
  - The interviewer will use this information to assess your proficiency in one or more job-related areas, which may include anything from adaptability to leadership to problem solving

# Beyond the Case: Preparing for the Fit Interview

- ▶ Know the Mission/Goals of the Firm (review the website in detail)
- ▶ Small Talk is vital - the walk to the interview. Be friendly and conversational throughout
- ▶ Know the skills they value as most important and in your answers explain when you used these skills and how those skills were critical to your success.
  - BCG - Communication, Analytics, Impact, Problem Solving, Teamwork, Leadership
  - McKinsey - Leadership, Problem Solving, Personal Impact, Achieving
- ▶ Be prepared to go in even more detail when prompted - layer your answers
- ▶ Be prepared for interruptions and changes in subject

# Beyond the Case: Interviewer Perspective

- ▶ What is the interviewer looking for:
  - Knowledge
  - Communication skills
  - General demeanor
- ▶ This behavior is representative of
  - Client / Consultant relationship
  - Credibility
  - Likeability

# Beyond the Case: Preparing for the Fit Interview

Sample behavioral-based interview questions:

- ▶ Tell me about yourself.
- ▶ Tell me about a time you made an impact?
- ▶ Tell me about a challenging situation you faced.
- ▶ Tell me about a time you needed to use your leadership skills.
- ▶ Would you return to your summer position?
- ▶ Why did you choose to pursue an MBA? Why Yale?
- ▶ Why consulting?
- ▶ Why this firm?
- ▶ Where do you see yourself in 10 years?
- ▶ Describe a situation in which you were able to use persuasion to successfully convince someone to see things your way.
- ▶ Describe a time when you were faced with a stressful situation that demonstrated your coping skills.
- ▶ Give me a specific example of a time when you used good judgment and logic in solving a problem.
- ▶ Give me an example of a time when you set a goal and were able to meet or achieve it.

# Beyond the Case: Preparing for the Fit Interview

- ▶ Tell me about a time when you had to use your presentation skills to influence someone's opinion.
- ▶ Give me a specific example of a time when you had to conform to a policy with which you did not agree.
- ▶ Please discuss an important written document you were required to complete.
- ▶ Tell me about a time when you had to go above and beyond the call of duty in order to get a job done.
- ▶ Tell me about a time when you had too many things to do and you were required to prioritize your tasks.
- ▶ Give me an example of a time when you had to make a split second decision.
- ▶ What is your typical way of dealing with conflict? Give me an example.
- ▶ Tell me about a time you were able to successfully deal with another person even when that individual may not have personally liked you (or vice versa).
- ▶ Tell me about a difficult decision you've made in the last year.
- ▶ Give me an example of a time when something you tried to accomplish and failed.
- ▶ Tell me about a recent situation in which you had to deal with a very upset customer or co-worker.
- ▶ Give me an example of a time when you motivated others.

# Beyond the Case: Preparing for the Fit Interview

- ▶ Tell me about a time when you delegated a project effectively.
- ▶ Give me an example of a time when you used your fact-finding skills to solve a problem.
- ▶ Tell me about a time when you missed an obvious solution to a problem.
- ▶ Describe a time when you anticipated potential problems and developed preventive measures.
- ▶ Tell me about a time when you were forced to make an unpopular decision.
- ▶ Please tell me about a time you had to fire a friend.
- ▶ Describe a time when you set your sights too high (or too low).



# Interview Preparation Worksheet

|                                                                                                     |                                                          |
|-----------------------------------------------------------------------------------------------------|----------------------------------------------------------|
| Interview preparation worksheet                                                                     |                                                          |
| What is my story of why I want to work in this field?                                               |                                                          |
| How did I develop an interest in this field?                                                        |                                                          |
| Why am I interested in this firm?                                                                   |                                                          |
| What headlines about my personality do I want to imply?<br>• ...<br>• ...<br>• ...                  | How will I communicate these?<br>• ...<br>• ...<br>• ... |
| What headlines about my analytical skills do I want to imply?<br>• ...<br>• ...<br>• ...            | How will I communicate these?<br>• ...<br>• ...<br>• ... |
| What headlines about my business sense and knowledge do I want to imply?<br>• ...<br>• ...<br>• ... | How will I communicate these?<br>• ...<br>• ...<br>• ... |
| What questions about the company will I ask at the end of the interview?                            |                                                          |

# Final Thoughts About Case Performance

- ▶ No matter what happens, keep your cool
- ▶ Even if your analysis is going horribly and the interviewer is content to let you struggle remain composed
  - Some interviewers will purposely try to rattle you, just to see how you react
  - Even if you are totally floundering, remain calm, take a moment to collect yourself, and attempt to get back on track
  - If they see you can handle pressure, you might still have a chance

# Crafting Your Story

- ▶ Interview & Networking 2-Minute story
- ▶ Resume
- ▶ Behavioral / Fit Interview
- ▶ Situation Opportunity Action Result (SOAR)

# Agenda

- ▶ Club Administrative Details
- ▶ Case Performance Discussion
- ▶ Beyond the Case
- ▶ Telling your story with the SOAR framework
- ▶ Case Practice

# Telling your story with the SOAR framework

- ▶ Your goal: impress your interviewer
- ▶ Interviewers want to hear that you can...
  - Prioritize
  - Be structured
  - Be concise
- ▶ Why use the SOAR framework?
  - Structures your story
  - Cuts the crap
  - Emphasizes leadership and results

# What does SOAR stand for?

- ▶ **Situation:** Introduce the story and provide context
- ▶ **Opportunity:** Describe the opportunity for leadership
- ▶ **Actions:** Summarize the actions you took
- ▶ **Results:** Summarize the results of your actions

# Dos and Don'ts

## ► Do:

- Be as brief as possible (aim for 2 minutes)
- Focus on your individual role in the story
- Use positive language (e.g. “Our challenge was” vs “We had to”)
- Make it interesting

## ► Don't

- Use stories without results or with negative results
- Include negative information or caveats
- Focus on the process, unless it's relevant
- Include irrelevant details

# Example of Unstructured Story

- ▶ This summer, Unilever held a team-building exercise where the interns had to compete in a cooking competition. It was called the Intern Cook-Off. The interns were split into teams of five and each team needed to come up with a menu for a three course meal and cook it. Afterwards, we would have to present our meal to a panel of judges. I suggested that we use the Dove “Go Fresh” line of products as our theme for the meal, since the line’s fragrances were mostly based on foods that we could use in the meal. I developed an original recipe for our main course called Energizing Chicken, and cooked it during the competition. I also presented the meal to the judges and they loved it! We didn’t win, but we got second place. It was a blast.



# Structuring Your Story

- ▶ Write/type your story and then pick out the key talking points for each part of the framework
- ▶ Memorize talking points, not a script
  - Not grammatically correct or complete sentences
- ▶ Bonus points for “headlining” your story with a super-quick description which encapsulates the main point
  - What is your message?
  - Take the opportunity to frame your story

# Example of Structured Talking Points

- ▶ **Headline:** Fun team-building exercise became leadership opportunity
- ▶ **Situation:** Unilever held intern cook-off. The challenge was to create three course meal, cook it, and present it
- ▶ **Opportunity:** Interns assigned teams of five. I Volunteered to lead
- ▶ **Actions:** suggested Go Fresh theme, created Energizing Chicken recipe, cooked main course, and presented to judges
- ▶ **Result:** Wowed the judges, won second place out of ten

# Agenda

- ▶ Club Administrative Details
- ▶ Case Performance Discussion
- ▶ Beyond the Case
- ▶ Telling your story with the SOAR framework
- ▶ Case Practice

# Management Consulting Case Interviews

Visit our website: <http://www.consultingcase101.com>

More consulting resources at: <http://www.consultingcase101.com/resources/>

Join Consulting Forum: <http://www.consultingcase101.com/forum/>

Our Blog: <http://www.consultingcase101.com/blog/>

Subscribe via RSS Feed: <http://www.consultingcase101.com/feed/>

Subscribe via Feedburner: <http://feeds.feedburner.com/consultingcase101>

Subscribe via Amazon Kindle: <http://www.amazon.com/Management-Consulting-Case-Interviews/dp/B0041HXS3Y/>

Follow us on Twitter: <http://twitter.com/consultingcase>

Follow us on Facebook: <http://www.facebook.com/pages/Management-Consulting-Case-Interviews/140533409321335/>